



Sanctuary™ 12K Energy Storage System (ESS)

8kW AC / 12kW PV Hybrid Inverter / Charger and 13.5kWh - 40.5kWh Lithium Iron Phosphate Battery





READ THIS INSTALLATION GUIDE & MANUAL IN ITS ENTIRETY BEFORE OPERATING THE UNIT.

This unit provides safe, silent, and renewable electric power. It is very important to carefully read this Installation Guide before using the product. Keep this guide for future reference.

Carefully read and comply with all safety directives. Otherwise, personal bodily injury or death may result.

Symbols Used:

-  **WARNING:** Indicates a hazardous situation which, if not avoided, could result in injury or death.
-  **CAUTION:** Indicates a hazardous situation which, if not avoided, could result in minor injury or damage to the equipment.
-  **NOTE:** Indicates an important step or tip that leads to best results, but is not safety or damage related.

Follow these directives for safe use:

-  **System Integrity:** The inverter and all system components must not be disassembled or altered by unauthorized personnel. Unauthorized modifications can compromise system safety, void warranties, and result in serious injury or death.
-  **Electrical Hazard:** Always disconnect all power sources, including the AC and DC terminals, before performing any maintenance or cleaning. Use lockout/tagout procedures to ensure the system remains de-energized during maintenance.
-  **Grounding Requirements:** Proper grounding is essential for safe operation. Ensure that the system is connected to a permanent, grounded wiring system, and comply with all applicable grounding and bonding requirements.

-  **Load Calculations:** Ensure that all load connections are made following the installation guide's specifications for wire size, breaker sizing, and load balancing. Incorrect wiring or undersized breakers may lead to system failure and hazardous conditions.
-  **Heavy Equipment:** This system includes heavy equipment. Use lifting assistance during installation to prevent injury.
-  **Compliance with Regulations:** All installation and maintenance activities must comply with local, state, and national electrical codes and standards, including but not limited to UL 9540 and UL 1973. Adherence to these standards is mandatory to ensure safety and compliance.
-  **Final Continuity Check:** Do not power on the system until a final continuity check is performed to ensure all connections are secure and correctly installed.
-  **Battery Handling:** Strictly follow the manufacturer's instructions for handling and installation of batteries. Failure to do so may result in fire, explosion, or electric shock.
-  **Tool Usage:** Exercise caution when using metal tools around batteries and electrical components. Dropping tools can cause short circuits, leading to potential explosions or fires.
-  **Ventilation:** The installation location must provide adequate ventilation to prevent overheating of the system components. Ensure that the unit is installed in a well-ventilated area according to the manufacturer's recommendations.
-  **Qualified Personnel Only:** Installation, service, and maintenance of this Sanctuary system must be performed by qualified personnel/technicians. Only certified professionals with the appropriate training should install or service the unit, with or without a battery.
-  **Manual Reference:** Refer to the full installation and operation manual for detailed instructions and safety guidelines. Failure to follow these instructions can result in personal injury, property damage, or voiding of the warranty.

General Safety Guidelines

-  **Emergency Procedures:** In case of fire, evacuate the area and contact emergency services immediately.
-  **Handling of Damaged Equipment:** If the unit or any of its components are damaged, do not operate the system. Contact a qualified technician, or ESS Support for inspection and repair.

SANCTUARY INSTALLATION GUIDE

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Lion Sanctuary Introduction

Congratulations on your purchase of the Lion Energy Sanctuary ESS (Energy Storage System)! By investing in this innovative system, you've taken a proactive step to ensure uninterrupted power for your home. The Sanctuary is more than just a backup; it's your reliable partner during unexpected outages.

Here's what you can expect from your Sanctuary:

Peace of Mind

Rest assured knowing that with the Sanctuary installed, you're prepared for any power disruption. Whether it's a storm, maintenance, or grid failure, your home stays powered. Say goodbye to noisy generators and the hassle of refueling. The Sanctuary silently takes over, keeping your essential appliances running smoothly.

Advanced Technology at Your Fingertips

The Sanctuary incorporates cutting-edge lithium iron phosphate battery cells, ensuring reliable performance. Monitor your system's performance, charge status, and energy usage conveniently using the app on your smartphone or PC.

Customizable Backup Solution

Tailor your backup power to your specific needs. Whether it's essentials like lights and Wi-Fi, or your entire home, the Sanctuary adapts to your requirements. Our proprietary software optimizes energy usage, ensuring efficient operation and cost-effectiveness.

Tax Credits for Clean Energy

Take advantage of available tax credits for installing an energy storage system in your area, making clean energy more affordable and accessible.

Expandable Options

Consider expanding your Sanctuary system to cover longer outages if you're in a higher-risk area. It's an investment in your family's safety and comfort. Adding more inverters or batteries to your system allows for greater power usage and storage capacity.

For more details and support, please refer to our official page on the back of this manual. Thank you for choosing Lion Energy—we're committed to keeping your home powered!

Commonly Used Terms

AC-Coupled Solar: Previously considered cutting-edge, AC-coupled solar has evolved over the years. In an AC-coupled system, solar panels generate DC power, which is then converted into AC power by a separate solar inverter or micro inverters. This AC power can be used to power home appliances, stored in batteries, or sold to the grid. The Lion Energy Sanctuary System offers the advantage of energy storage, allowing you to utilize the power you produce.

AHJ (Authority Having Jurisdiction): This organization, agency, or inspector is responsible for making sure your installation meets applicable code requirements.

Bypass: Bypass occurs when essential or backup loads are powered by the grid rather than the Sanctuary system. It serves as a temporary state during system maintenance, similar to taking a detour in a construction zone. A bypass switch allows homeowners or installers to temporarily bypass the Sanctuary system for servicing or troubleshooting while still maintaining power.

C (in relation to battery charge/discharge rate): The C rate for batteries is related to the Amp-hour capacity. For example, if a battery capacity is rated at 280Ah, then one C rate is 280 Amps, and 0.1C is 28A.

Consumption: Consumption refers to the power used by electrical loads or the entire home. CTs track consumption patterns, providing insights into energy usage.

CT (Current Transformer): CTs are devices used to measure the current flow in another circuit. In the Lion Energy System, external CTs are installed according to specific guidelines to accurately monitor current flow.

DC: DC refers to direct current, such as solar power and battery output.

Essential/Backup Loads: Essential or backup loads refer to the circuits and devices powered by the Lion Energy Sanctuary system. These loads are backed up by batteries, inverters, and solar power, ensuring uninterrupted power supply during outages. Excess solar power can offset the usage of non-essential loads, providing the benefits of self-consumption and energy savings.

Grid: The grid refers to the power supply from the utility company. Energy meters installed by utility companies measure the amount of power consumed by a home, typically billed monthly.

Integration with Existing Solar: The Lion Energy Sanctuary seamlessly integrates with almost any existing solar system on the market, ensuring compatibility and flexibility.

Load: A load refers to any active electrical circuit or device. Essential or backup loads are powered by the Sanctuary system.

MPPT: Maximum power point tracking. This is a type of power converter that is able to extract the maximum solar power available.

PV (Photovoltaic): PV refers to solar energy generation using photovoltaic cells. PV systems convert sunlight into electricity, contributing to sustainable power sources. The Sanctuary system works in conjunction with PV systems to store and distribute solar power efficiently.

Sanctuary ESS Overview

The Sanctuary is a multi-functional Energy Storage System (ESS), which incorporates the functions of an inverter, solar charger, battery charger, generator (not included), and lithium iron phosphate battery. The Sanctuary is commissioned and monitored by way of Lion App, available on smart phone and PC.

The Sanctuary ESS can be considered your own personal power plant. Whether you are covering essential loads or covering an entire home, the correct sizing of the Sanctuary is key for the system to function properly. Before sizing, you must review the Sanctuary technical specifications.

To size the system, consider the loads you would like the Sanctuary to cover. Typical essential loads are refrigeration, furnace, lights, Internet, and garage door.

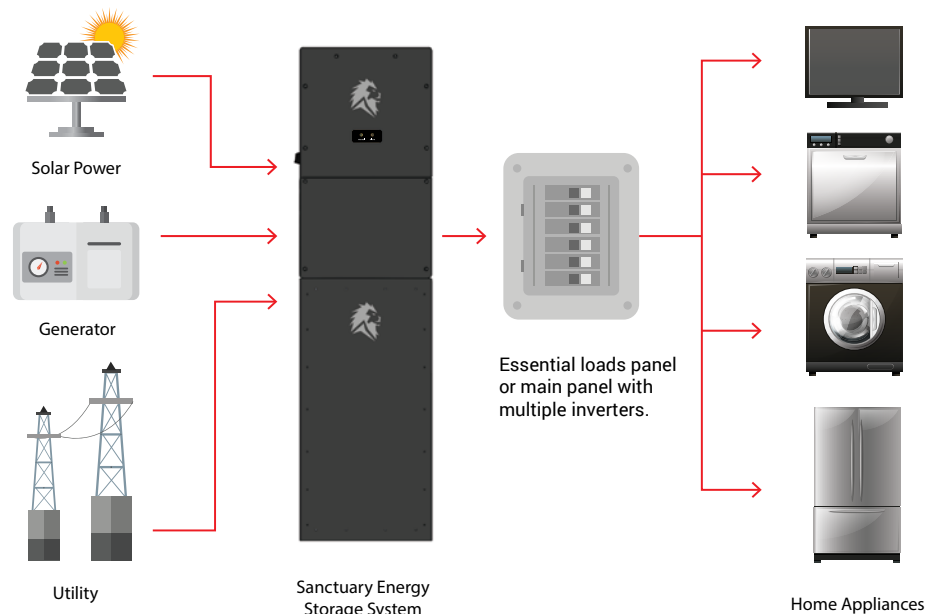
Review your existing load center and identify the breaker size for each of the loads you would like to cover. Note - any load larger than a 30 amp breaker will require multiple inverters to cover. Using the code compliant load calculation method for your jurisdiction, determine the number of continuous amps required to cover the desired loads. Each Sanctuary system can provide 33.3A AC off-grid (stand-alone). Any requirement above this will need additional Sanctuary inverters placed in parallel. Please contact Lion Energy for more details.

Note that the Sanctuary is only sold with both the inverter and battery as a system as both are required for full functionality of the unit. If paired with solar, the recommend minimum PV input is 3kW per battery of energy storage (13.5kWh each).

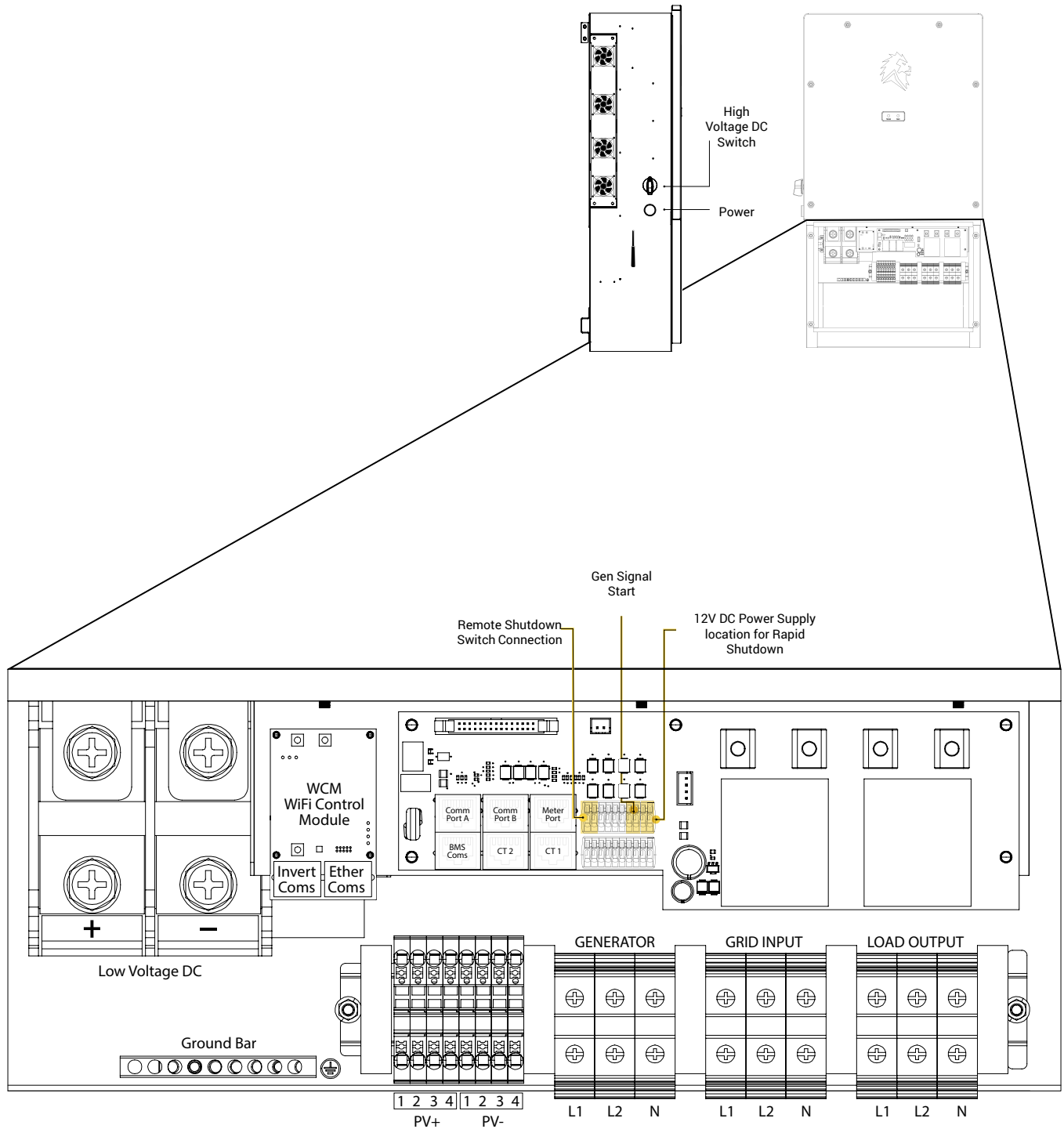
Basic System Architecture

This figure depicts the basic application of the Lion Sanctuary System. Power is fed into the system from the power grid, solar power array or generator to have a complete running system.

The Lion Sanctuary Energy Storage System can provide power for residences, including appliances, communication equipment, lights and other devices.



Inverter Overview



Inverter Parts and Components



A black rectangular inverter unit with a silver lion logo in the center. Below the logo is a small control panel with two indicator lights labeled 'Normal' and 'Alarm'. The unit is secured with screws around the perimeter.

12K Inverter
(#50170168)

1 Per Inverter



Two black metal L-shaped brackets with pre-drilled holes for mounting.

Inverter Standoff Bracket Set

1 Per Inverter



A long, thin black metal bar with four screws pre-installed at regular intervals.

Inverter Wall Mount

1 Per Inverter



Two black cables with white plastic CT (Current Transformer) sensors attached to the ends.

CT Pair

1 Per Inverter



Two black metal brackets with four screws each, designed for wall mounting.

Inverter Wall mount Bracket Set
(#50170168)

2 Per Inverter



A coiled black Cat5 Ethernet cable with RJ45 connectors on both ends.

Cat5 Cable (10')
(#90000136)

1 Per Inverter



A black cable with metal eyelet connectors at both ends.

AWG2 Black Eylet-Eyelet Cable (24")
(#90000516)

1 Per Inverter



A red cable with metal eyelet connectors at both ends.

AWG2 Red Eylet-Eyelet Cable (24")
(#90000517)

1 Per Inverter



Two black busbar units with multiple screw terminals for electrical connections.

Busbar Set
(#90000508)

Battery Parts and Components



13.5kWh Battery
(#50170132)

1 Per Battery



Battery Bracket

1 Per Battery



Freestanding Wire Box
(#90000504)

1 Per Battery



Wire Box Cover
(#90000506)

4 Per Battery



Wire Box Housing Bolts
(#90000518)

1 Per Battery



**AWG2 Battery
Cable Red (55")**
(#90000501)

1 Per Battery



**AWG2 Battery
Cable Black (55")**
(#90000500)

1 Per Battery



Battery COM Cable
(#90000508)

1 Per Battery

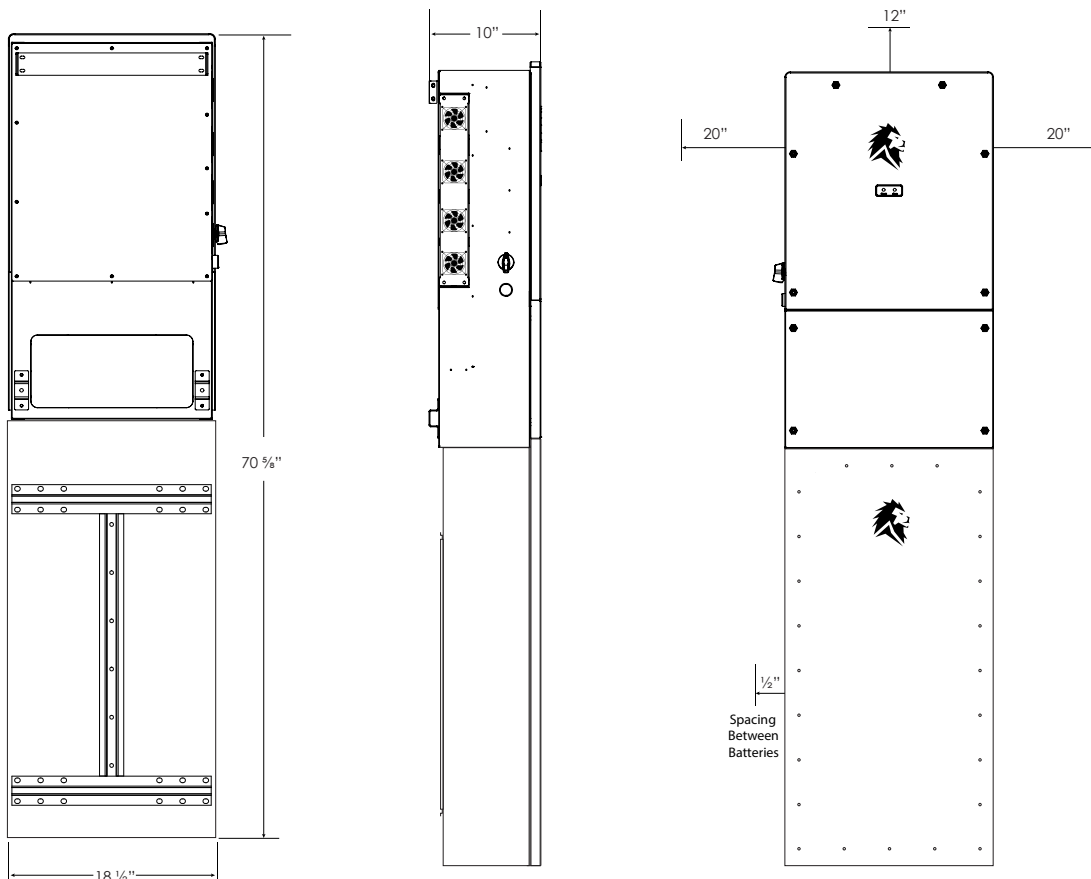


RJ45 4-Port Splitter
(#90000509)

Installation Location

Before installing the Lion Sanctuary System (inverter/charger and battery), consider the following when choosing a location for installation:

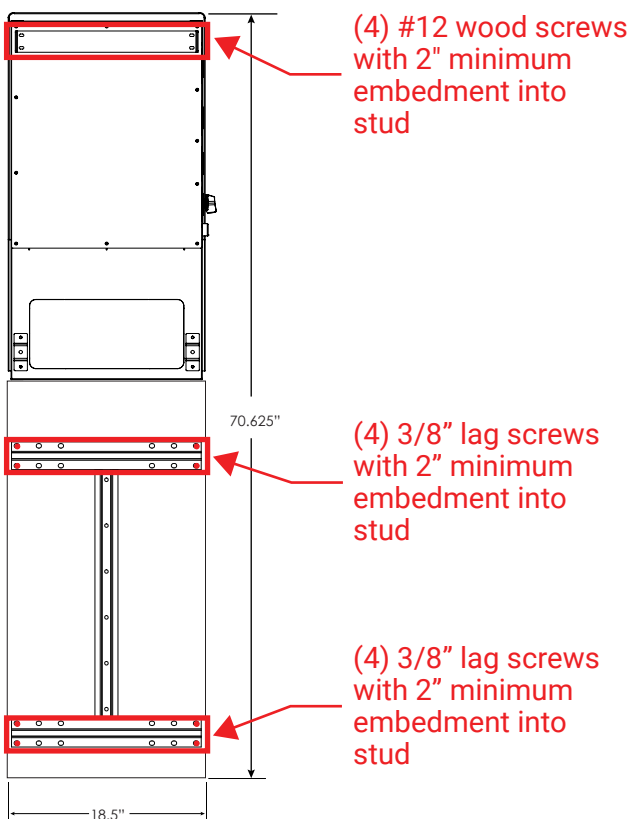
- Install the Sanctuary System in a climate controlled location, regulated temperature between 32° to 86° F. The Sanctuary ESS cannot be installed in a living space. In other words, the Sanctuary ESS needs to be installed in a garage or utility/storage room, not your family room.
- Be sure to keep other objects and surfaces away from the unit to permit adequate heat dissipation and provide space for wiring access. For proper air circulation, provide a clearance of at least 20 inches to the side and at least 12 inches above the inverter. The batteries have been tested at 4" apart and approval was given to install with a minimum of 0.5" between batteries. However, the AHJ may require 4" minimum spacing between batteries. Check with your AHJ for their requirements.
- To reduce installation costs, it is recommended to install the Sanctuary near existing electrical panels when possible.
- Note that the inverter, fans, and other internal components emit sound at 38dB (slightly louder than a standard computer fan).
- For proper air circulation and heat dissipation, a clearance of at least 36" in front of the unit is required for ventilation and service access.
- We recommend not installing the DC solar switch higher than 72" from the ground.
- The battery and inverter must be mounted level. Therefore, a 90° angle is necessary for the proper installation of the Sanctuary System.



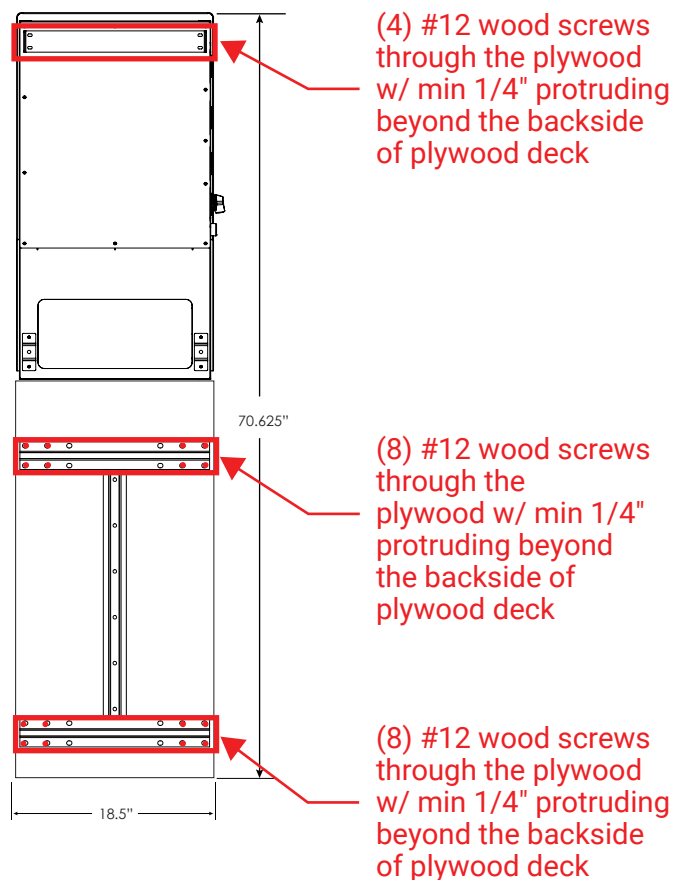
Fastener Requirements

Minimum fasteners per equipment

Fasteners into stud



Fasteners into plywood

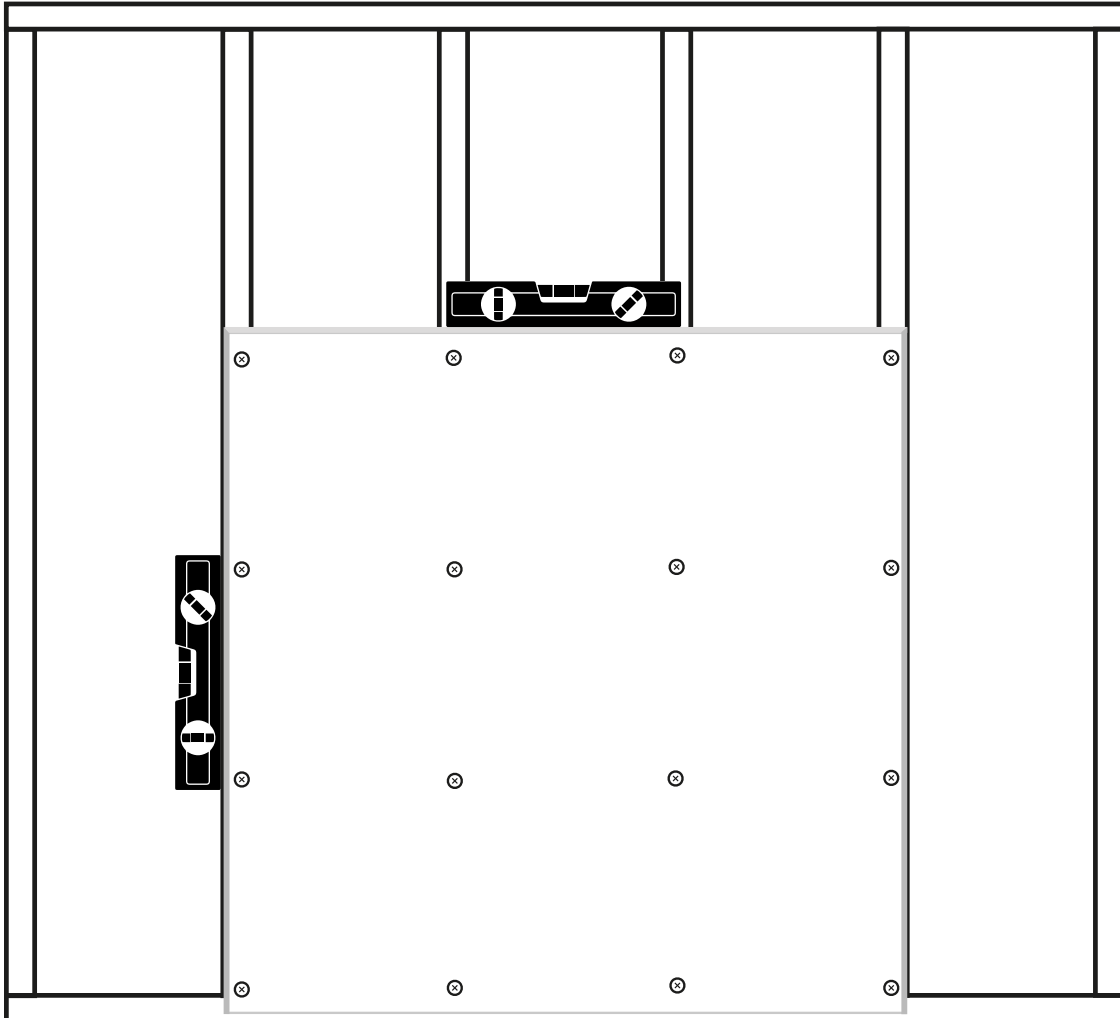


i Indoor Only

1. Installation of the lion energy's sanctuary battery system shall be limited to indoor use only.
2. For installation to plywood, minimum 1/2" plywood must be installed.

Location Preparation:

The Lion Sanctuary ESS battery is designed to be mounted on a wall using a French Cleat system. The inverter is designed to be mounted to the top of the battery and directly to the wall. It is recommended that you install a suitable non-flammable surface onto an existing wall for anchoring. Before mounting the brackets, ensure the surface is square and level.



After confirming that the wall surface is non-flammable and securely supported—such as by anchoring directly to studs or approved fire-rated material—the next step is to install the mounting brackets.



Ignition Hazard

Do not install the system in an environment with flammable vapors, such as gasoline, as sparks from relays may ignite vapors

Mounting Brackets

Step 1: Mount Inverter standoff brackets and Inverter wall mount brackets to the back of the inverter.

Step 2: Measure a minimum of 30½" from the floor to the top of the battery bracket and secure the battery bracket to the wall.

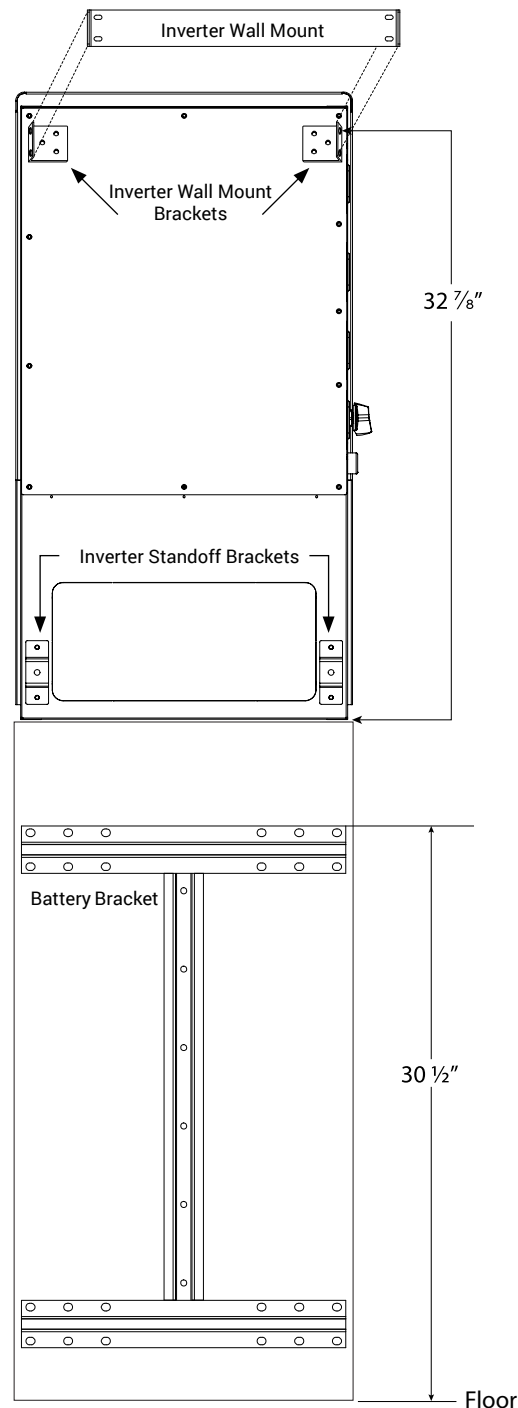
Step 3: Mount battery on the battery bracket.

Step 4: Measure 32 7/8" from the top of the battery to the top of the inverter wall-mount bracket. Mark placement and secure inverter wall mount.

Step 5: Mount inverter

Additional batteries should be mounted at the same height as mounted in step 1 and 2.

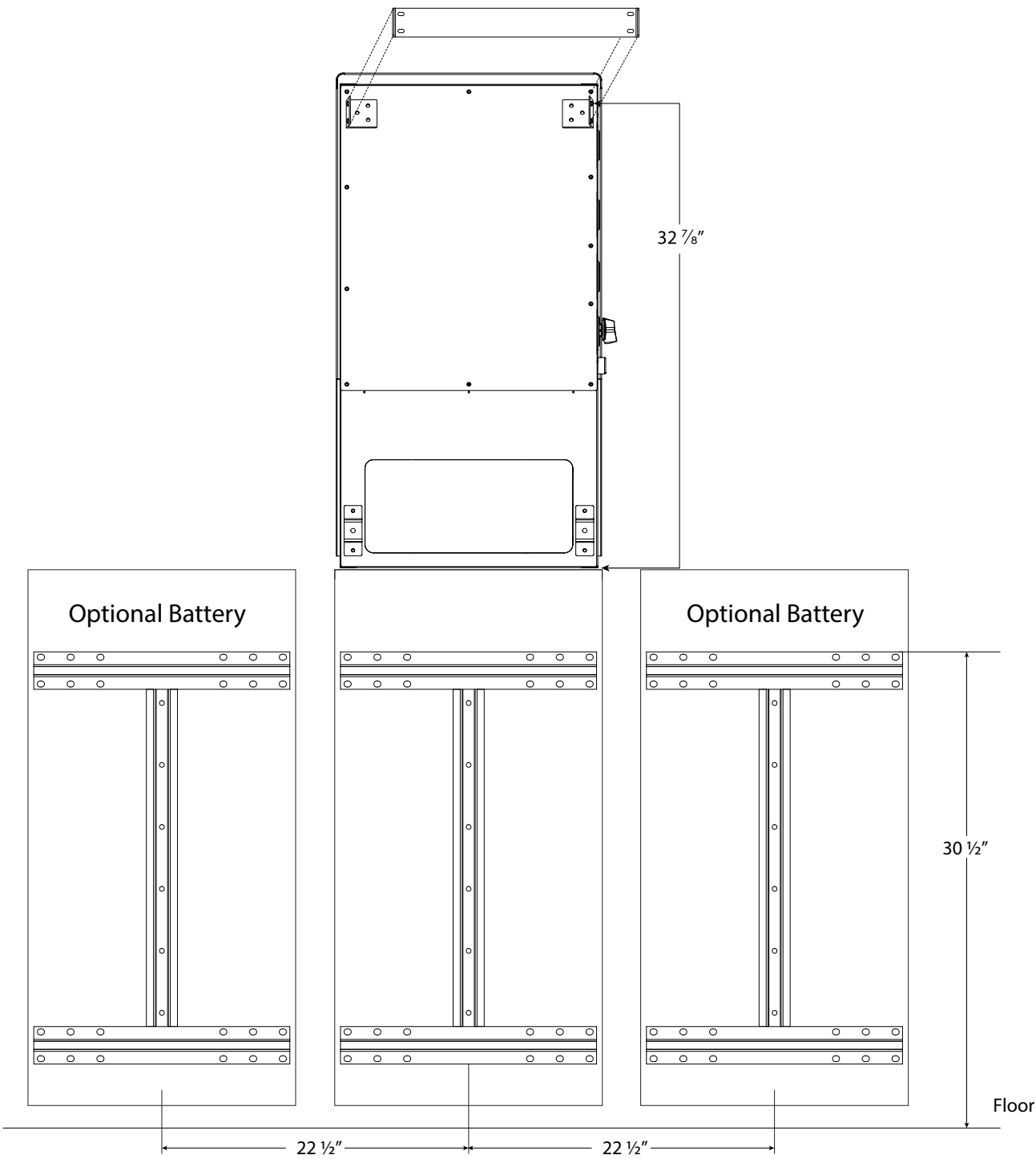
If mounting multiple inverters follow the spacing guidelines as previously mentioned



Multiple Inverter Setup

To provide proper airflow you must space the **inverters 20"** from each other.

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(Yields a recommended 4" spacing between batteries for ease of installation)

Minimum battery spacing allowable is 4 inches as stated on UL Certificate.

Battery Voltage Check

Before connecting batteries in parallel, they should be within 0.5V of each other. If the batteries are not within 0.5V of each other, excessively high current may flow between batteries when they are connected in parallel.

If any battery is above 53.5V resting voltage at room temperature, it is probably above 98% charged. It can be quickly discharged down to 53.5V by running the inverter using battery power as its only power source.

If any battery is lower than 51.2V resting voltage at room temperature, it is probably lower than 7% charged. It should be charged up to within 0.5V of the other batteries before connecting them in parallel.

The following steps can be completed after commissioning the system.

Step 1. Before connecting the batteries, start with only the negative power cable connected to each battery, with the positive cable on each battery remaining unplugged. Note: This may trigger a battery disconnected alarm.

Step 2. Temporarily change these settings:

- For a single inverter system, set “inverter max charge current” to no more than 140A.
- For parallel inverter systems, set “system charge current” to no more than 140A.
- If there is not enough solar power available, set the inverters to battery priority mode so that the batteries will charge using grid power.

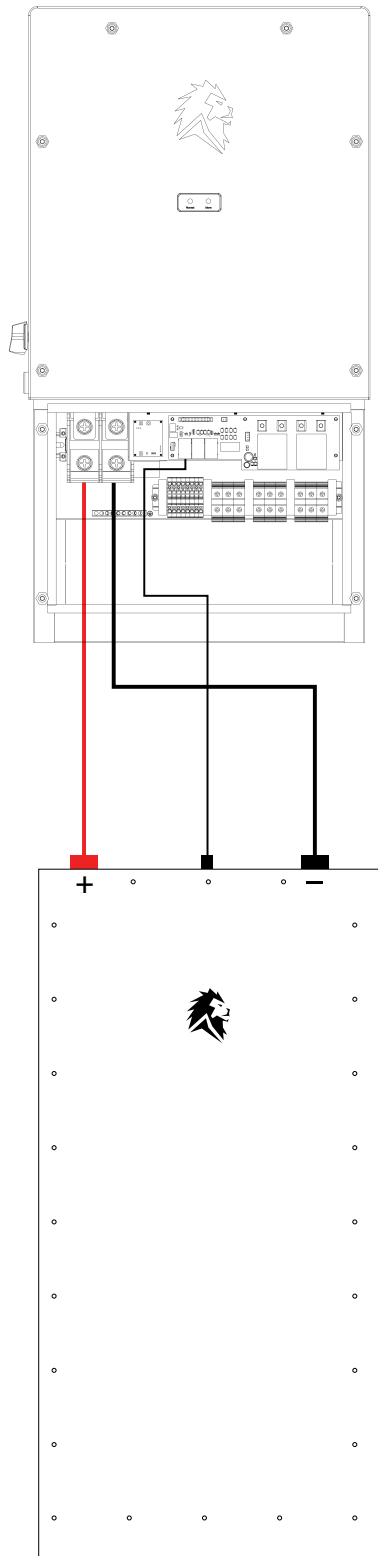
Step 3. Check the voltage on the inverter’s battery terminals. If it’s below 40V, use the battery awaken function in the web app. After a minute, the voltage should rise above 50V. Then plug in the positive battery cable for the lowest voltage battery.

Step 4. After the lowest battery is charged up to within 0.5V of the next lowest battery, plug in the positive cable of the next lowest battery.

Step 5. Continue charging and repeat step 4 until all batteries are plugged in.

Step 6. Change the “system charge current” back to the original setting. Emergency mode can be turned off.

Low Voltage DC Wiring: 1 Inverter



When wiring 1 inverter to 1 battery,

first check the voltage of the Battery. It should be between 45-55.6VDC. If it is not, contact LionESS support at (435) 244-3352.

Connect the battery cables to the inverter **first**. Now connect the cables to the battery receptacles last.

Next, connect the BMS communication cable to the RJ45 port labeled BMS Coms on the inverter and then connect the 4-pin aviation connector to the battery.

When wiring 1 inverter to multiple batteries,

Step 1: Connect the provided eyelet to eyelet battery cables to the inverter first, then to the provided busbars.

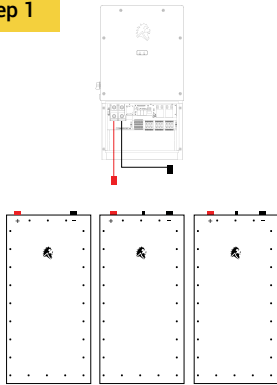
Step 2: Using the provided battery cables, connect all negative cables from the negative busbar to the negative battery receptacles,

Step 3: Now connect the positive cables from the positive busbar to the positive battery receptacles.

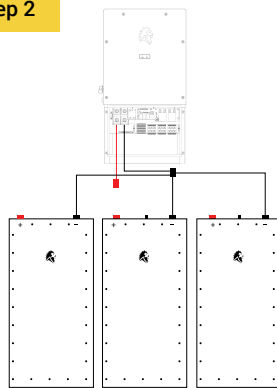
Step 4: Next, connect the provided BMS communication cable splitter to the RJ45 port labeled BMS Coms on the inverter. Connect each battery BMS communication cable to each 4-pin aviation connector on each battery.

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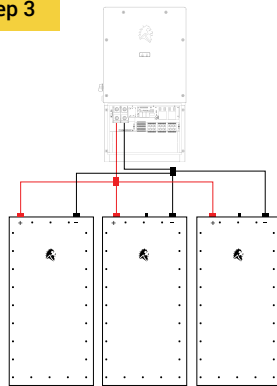
Step 1



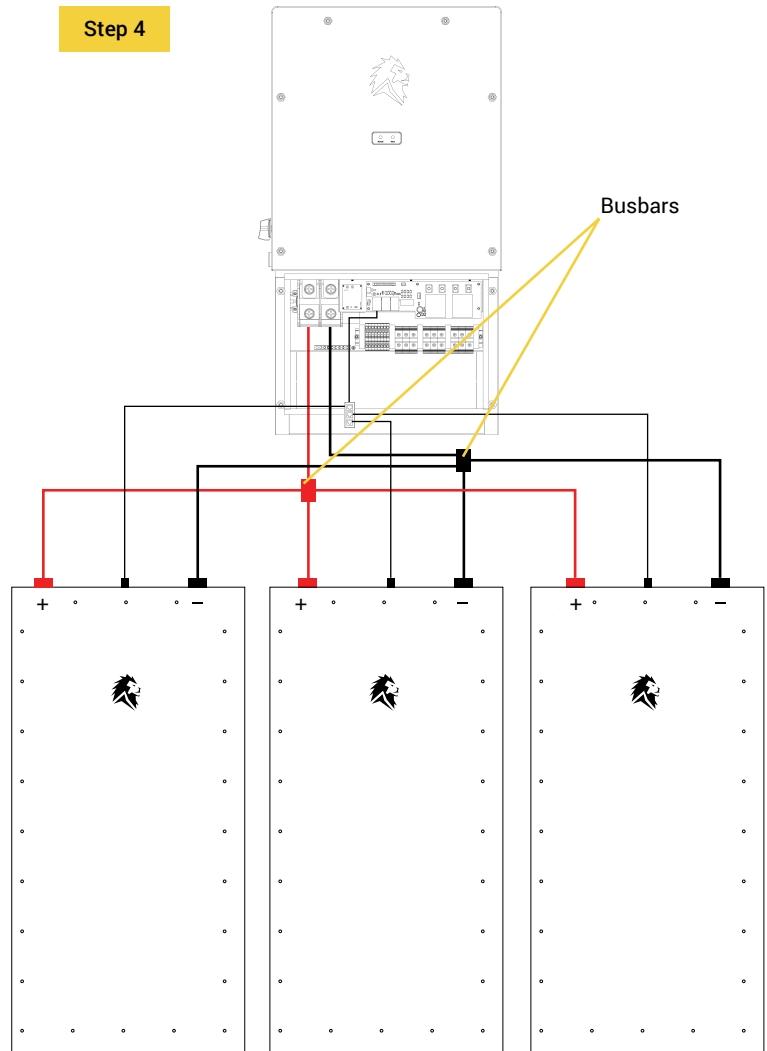
Step 2



Step 3



Step 4



Battery Voltage

When connecting multiple batteries, check voltage on each battery first. Batteries must be within 0.5V in order to connect in parallel.

Low Voltage DC Wiring: Multiple Inverter

(3 Inverters, 6 Batteries)

Step 1: Mount busbars directly below each inverter. Connect busbars to each other.

Step 2: Using the provided eyelet to eyelet cables connect each inverter to their corresponding busbar.

Step 3: Connect each negative battery cable eyelet end to its corresponding negative busbar. Do not connect the cables to the battery terminals yet.

Step 4: Connect each positive battery cable eyelet end to its corresponding positive busbar.

Step 5: Plug in the negative (-) battery cables to the negative battery terminals.

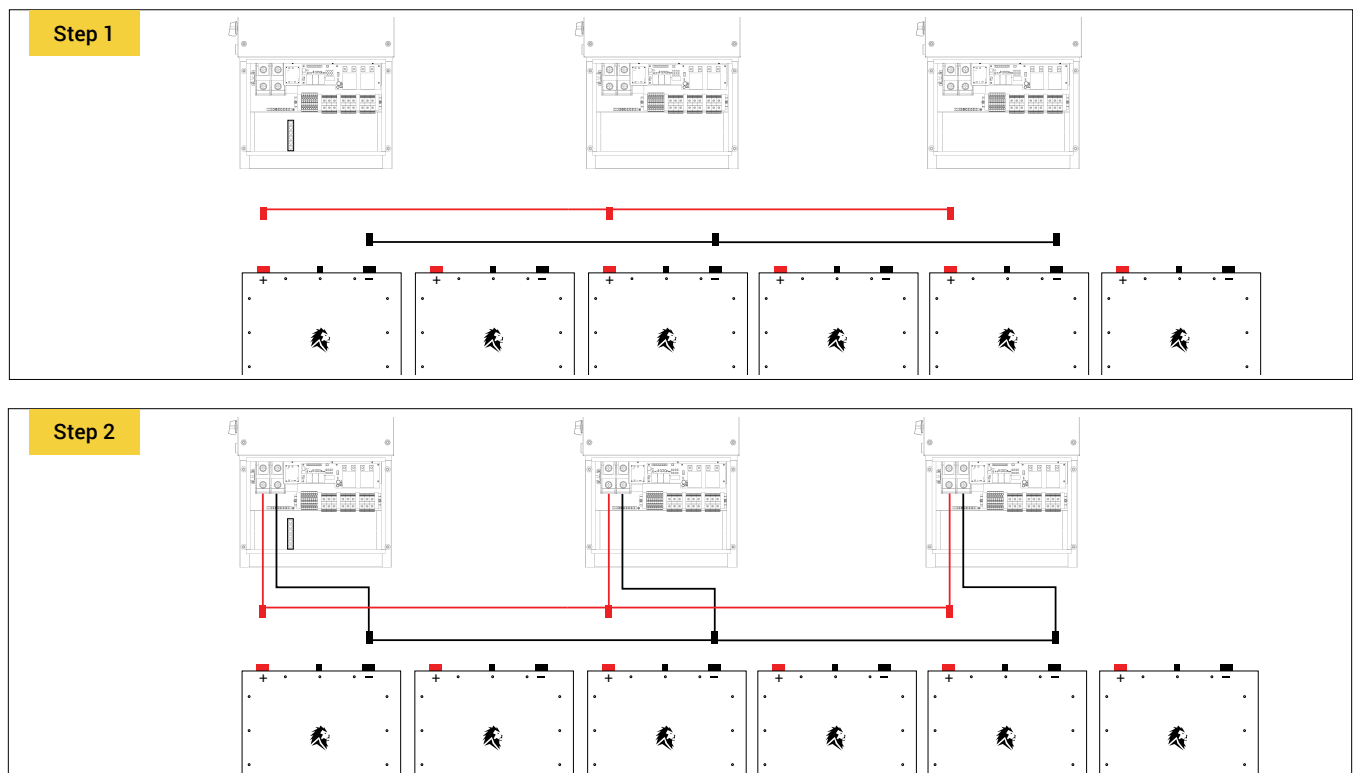
Step 6: Plug in the positive (+) battery cables to the positive battery terminals.

Step 7: Connect the BMS Com Splitter to the bottom left RJ45 port labeled BMS Coms on the Parent Inverter. Next, connect each battery BMS Coms cable to the BMS Com Splitter. Lastly, connect the 4-pin Aviation end to each battery.

Battery Voltage

When connecting multiple batteries, check voltage on each battery first. Batteries must be within 0.5V in order to connect in parallel.

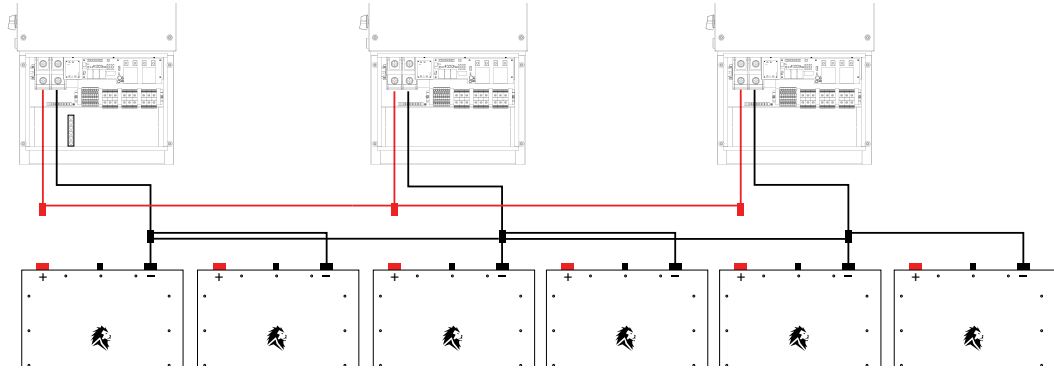
When connecting busbars battery cables must be the same length throughout the entire system and have a correct degree and voltage rating.



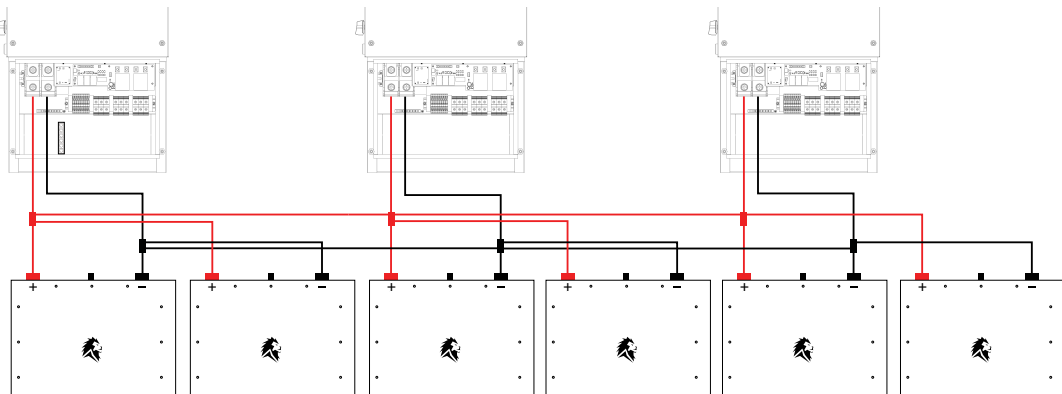
For more detailed instructions on connecting the communication cables between batteries see page 30-31

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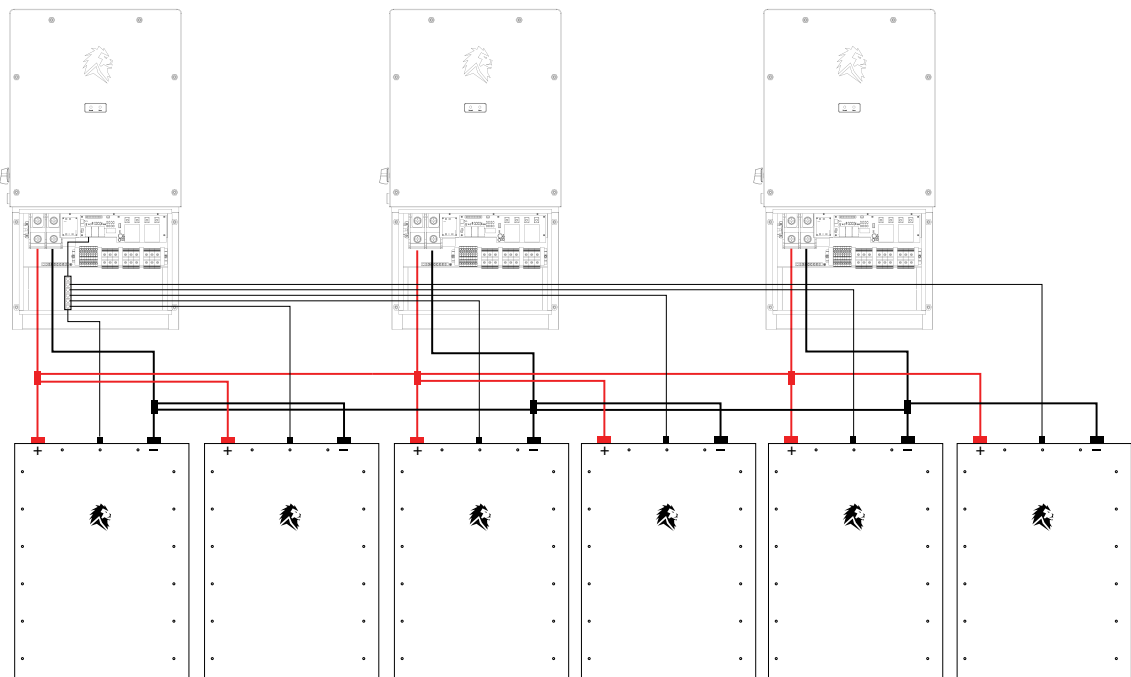
Step 3



Step 4

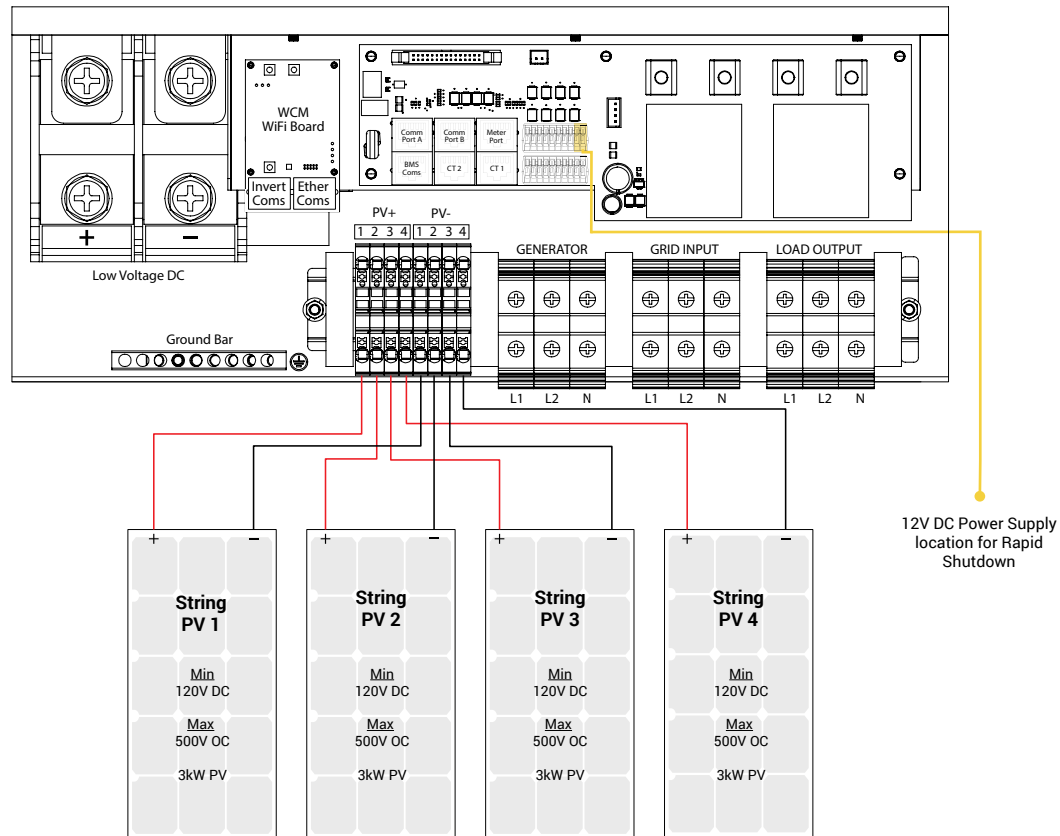


Step 5



High Voltage DC Wiring

The Sanctuary Inverter has 4 MPPTs that are capable of 3kW of solar each, for a total of 12kW. The minimum usable voltage is 120V DC. The maximum allowable voltage is 500VOC. The max PV ISC is 15A DC per MPPT. When configuring your PV strings, you must account for the panel wattage rating and voltage. Refer to the Sanctuary technical specifications for further detail.



Correctly connect positive and negative

Before making the final DC connection or turning on the high-voltage DC switch / disconnect, be sure the positive (+) connects to positive (+) and negative (-) connects to negative (-).



Professional wiring installation

All wiring must be performed by a licensed electrician.



High voltage

Make connections while the unit is powered off. The PV(-) terminals are typically at -240V DC while the unit is in operation.

PV Connection

It is important for system safety and efficient operation to use the appropriate cables for the PV module connections. To reduce risk of injury, use the recommended cable size. The Sanctuary system accepts up to 10 AWG wire for PV connection.

PV Module Wire Connection

Verify correct polarity of all wire connections for the PV modules and PV input connectors. Insert the wires and connect the positive pole (+) of the connection wire to the positive pole (+) of the PV input connector. Connect the negative pole (-) of the connection wire to the negative pole (-) of the PV input connector.



Avoid PV modules with current leakage

To avoid any malfunction, do not connect any PV modules with possible ground current leakage to the inverter. Only the PV racking is grounded. The inverter's PV wiring must not have any path to ground.

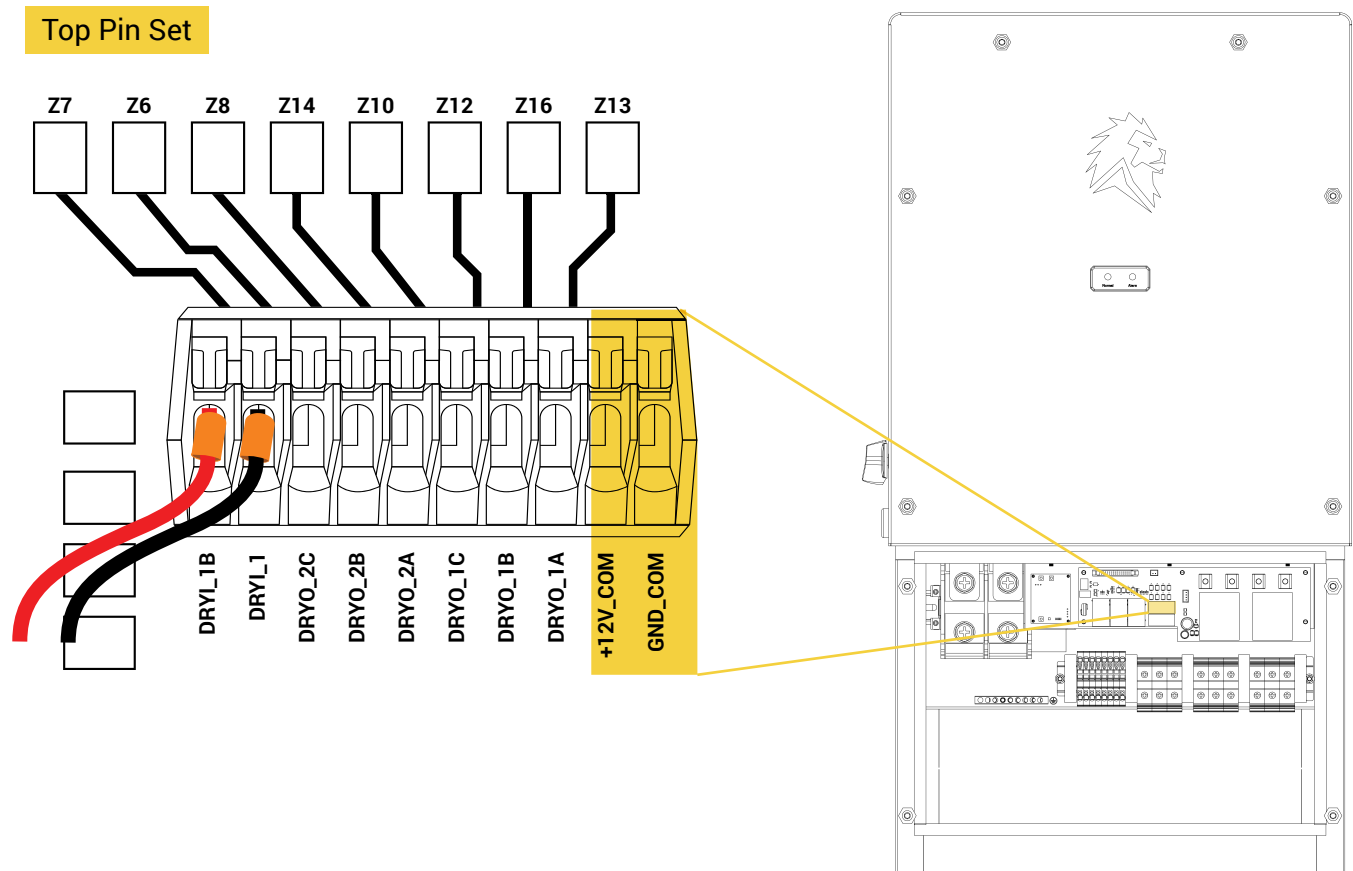
Rapid Shutdown

The national electric code (NEC) requires the use of a rapid shutdown system for PV systems on buildings. This is for first responder safety during rooftop ventilation or rescue operations. See NEC, section 690.12 for more information.

The Sanctuary inverter has a built in 12 Volt DC, 1A power supply for a rapid shutdown (RS) transmitter. The RS transmitter is polarity sensitive.

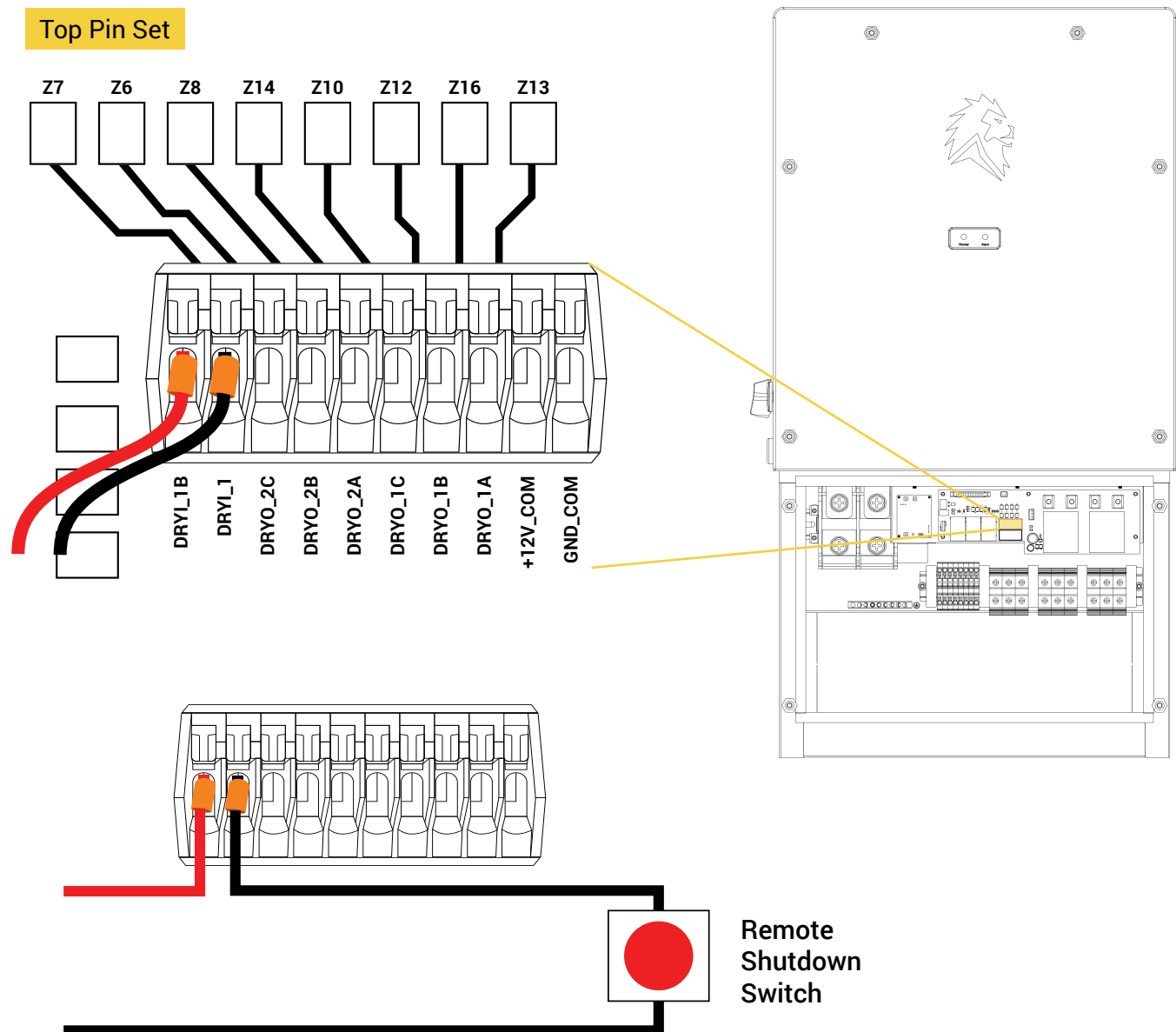
The 12 Volt supply is located as highlighted below on the pins labeled +12V_COM and GND_COM.

When either the remote shutdown switch is pressed (open circuit), or the power button is turned off, the inverters will turn off, including this 12V power source.



Remote Shutdown Switch

If you are installing a remote shutdown switch, remove the black wire on the 12 volt circuit board on the inverter. Next, mate this wire and run one wire out to the remote shutdown switch. Then on the other side of the switch, run one wire back and connect it into the contact where the black wire used to be on the circuit board.



AC Wiring: 1 Inverter

For grid-tied installation, the Sanctuary is capable of up to 90 Amp grid-supplied pass-through. The grid input power must come from an appropriately sized circuit breaker. The maximum wire size the inverter terminals can accept is 2 AWG copper.

Inverter to cover Essential Loads

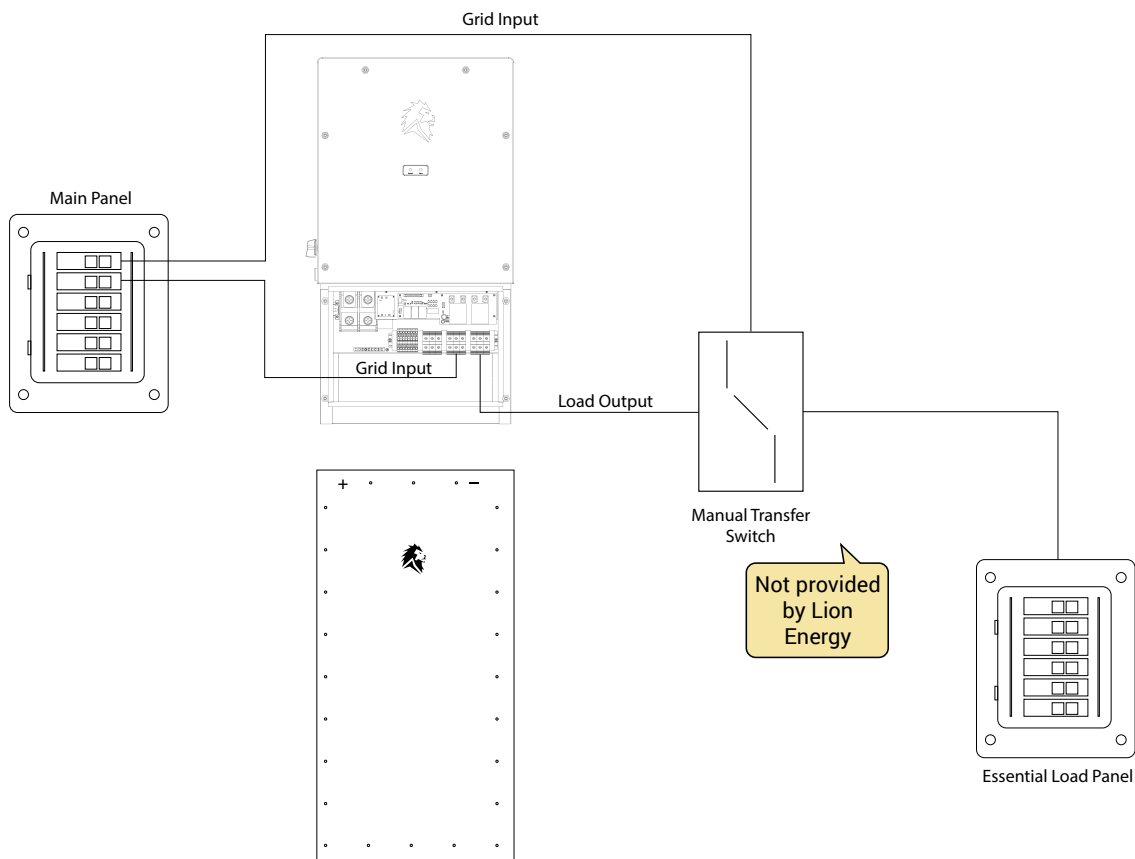
Install two breakers into the Main Panel.

One breaker will be the grid input for the Sanctuary Inverter. The load output of the Sanctuary Inverter will go to a 3-position manual transfer switch. The next breaker will go from the Main Panel directly to the 3-position manual transfer switch. From the manual transfer switch, lines will go directly into the Essential Loads Panel.



3-Position Switch

A 3-position manual transfer switch is recommended but not required.



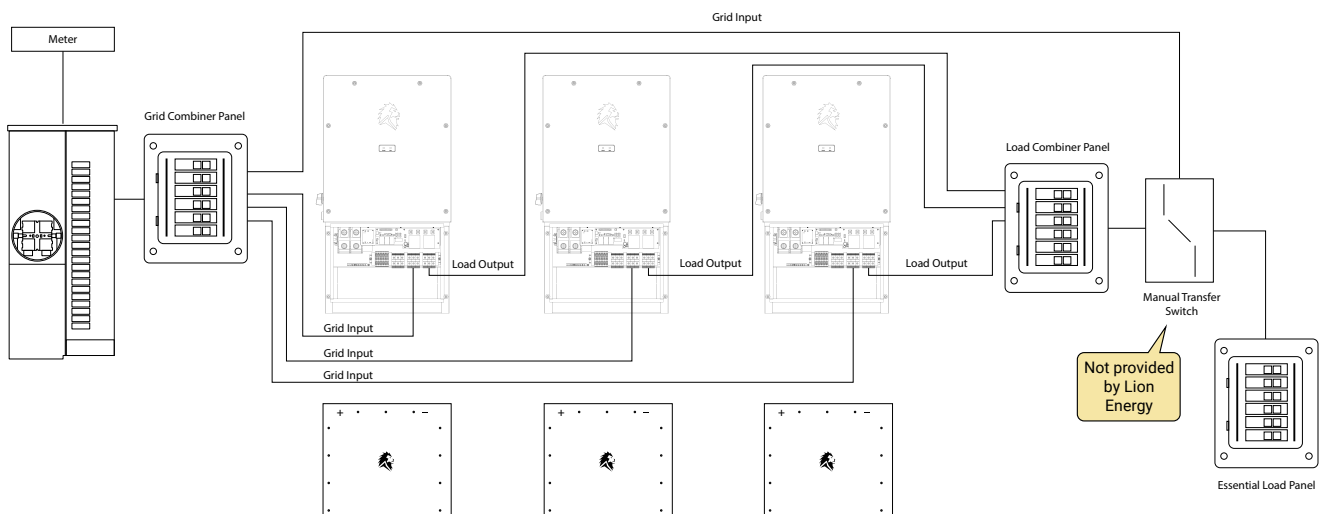
AC Wiring: Multiple Inverter

For a Grid-Tied installation, the Sanctuary is capable of up to a 90 Amp grid-supplied pass-through. The grid input power must come from an appropriately sized circuit breaker. The maximum wire size the inverter terminals can accept is 2 AWG copper.

3 Inverters to cover whole home backup

Between the Meter and the existing Loads Panel, install a grid combiner panel with 4 breakers.

Three breakers will be the grid inputs for each of the Sanctuary Inverters. The fourth breaker will go from the grid combiner panel directly to the 3-position manual transfer switch. The load output of each Sanctuary Inverter will go to a load combiner panel. Then from the combiner panel to the 3-position manual transfer switch. From the manual transfer switch, lines will go directly into the Essential Loads Panel.



L1 L2 PHASING

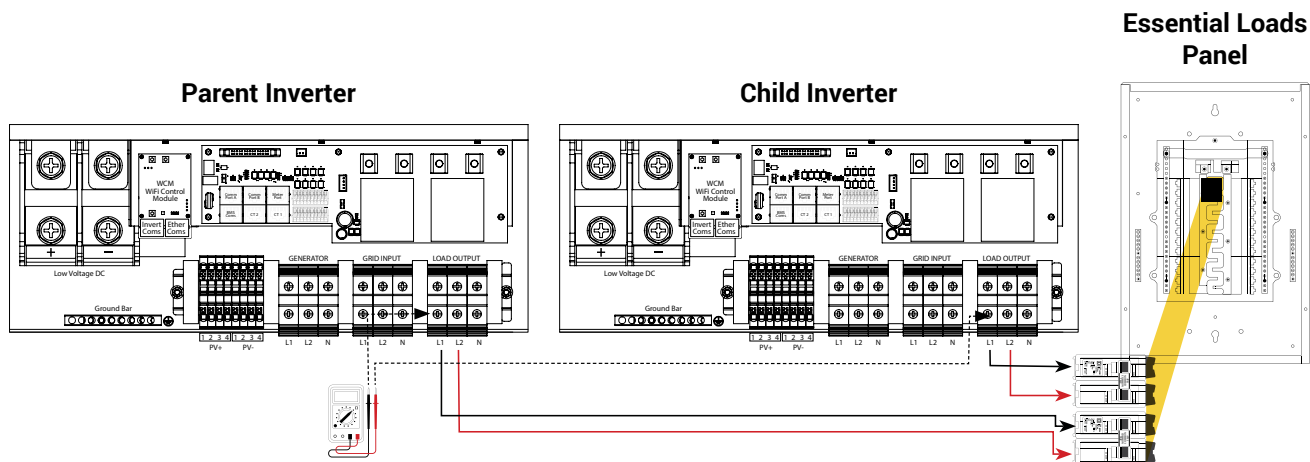
It is of vital importance that the grid input and load output phases on multiple breakers match. If a phase is crossed, that will result in a short circuit from line1 to line2. Crossing phases WILL cause a catastrophic failure on the equipment. See continuity testing on page 28.



Breaker Sizing

The three breakers feeding the sanctuary system from the main panel need to be the same amp rating with a maximum of 90A each. The fourth breaker that directly feeds the manual bypass should be roughly equal to the total of the other three breakers. For (example 3 breakers @ 70A ea to feed the sanctuary units and a 200A breaker to go to the manual transfer panel.)

Continuity Testing: Multiple Inverters



When installing a multiple inverter Sanctuary, it is essential to not cross phases. This means, Line 1 needs to remain Line 1 on Grid Input and Load Output for all Inverters. Line 2 needs to remain Line 2 on Grid Input and Load Output for all Inverters. Neutral needs to remain the same as well. There are three ways phases can be crossed. Physical wiring, the physical bussing of the service panel, and the breaker itself. A simple way to ensure phases have not been crossed is to conduct a continuity test.

When installing multiple inverters, it is **REQUIRED** that you individually test the phasing on the grid input, generator input, and load output terminals of each inverter.

Grid Input Continuity Testing

If there is no user-accessible AC disconnect above the inverter grid breakers, use caution when making measurements on live circuits.

- Step 1.** Make sure the AC power button on each inverter is turned off.
- Step 2.** Turn off the load breakers in the load combiner panel so that the load port on each of the inverters is disconnected from the other inverters.
- Step 3.** Turn off the power feeding the grid breakers for the Sanctuary inverters, but leave the inverter grid breakers in the on position. If there is no disconnect above the inverters' grid breakers, measurements will be done on live wires.
- Step 4.** Measure voltage between grid L1 on the first inverter and grid L1 on the next inverter. It should read zero volts. Check continuity between grid L1 on the first inverter and grid L1 on the second inverter. These should be close to zero ohms. Measure voltage between grid L1 on the second inverter to grid L1 on the third inverter. It should read zero volts. Grid L1 on the second inverter to grid L1 on the third inverter should also be close to zero ohms.
- Step 5.** Measure voltage between L2 on the first inverter and L2 on the next inverter. It should read zero volts. Check continuity between L2 on the first inverter and L2 on the second inverter. These should be close to zero ohms. Measure voltage on L2 from the second inverter to L2 on the third inverter. It should read zero volts. L2 on the second to L2 on the third inverter should also be close to zero ohms.
- Step 6.** Measure voltage between L1 and L2 on each inverter. If grid power is present, it should read 240V on each inverter. If grid power is not present and no voltage is detected, there should be no continuity between L1 and L2. Do not measure continuity across 240V.
- Step 7.** On each inverter, measure voltage between grid neutral and ground. It should read zero volts. Measure continuity between grid neutral and ground. It should be close to zero ohms due to the neutral to ground bonding in the first panel after the utility meter.

Load Output Continuity Testing

Load Output Continuity Testing

After confirming that the inverters have all power sources turned off, there should be no voltage present at the grid input, generator input, or load output terminals. Start by checking the load combiner panel to ensure there is no voltage present on the panel. (Note: If voltage is present and the inverters are turned off, there may be another power source feeding the panel, such as a mis-wired bypass switch.) After confirming the panel is not live, you can now turn on the load output breakers from each inverter. This allows the circuit between the inverters to be closed so you will be able to check continuity of phases between each inverter.

If you installed a manual bypass switch, switch the essential loads panel to the grid power position. This will disconnect the loads from the inverter load breakers. If there is not a manual bypass switch, turn off all the breakers in the essential loads panel for this test. Leave the inverters' load output breakers on so that each inverter's load port is connected to the other inverters' load ports.

With the inverters off, there should be no voltage present on the load terminals of any inverter.

Load Output Continuity Testing: L1

Measure continuity between load L1 of the parent and load L1 of the child inverter. It should be close to zero ohms. Measure continuity between load L1 of the child inverter and load L1 of the next child inverter. It should also be close to zero ohms.

Load Output Continuity Testing: L2

Measure continuity between load L2 of the parent and load L2 of the child inverter. It should be close to zero ohms. Measure continuity between load L2 of the child inverter and load L2 of the next child inverter. It should also be close to zero ohms.

Load Output Short Circuit Check

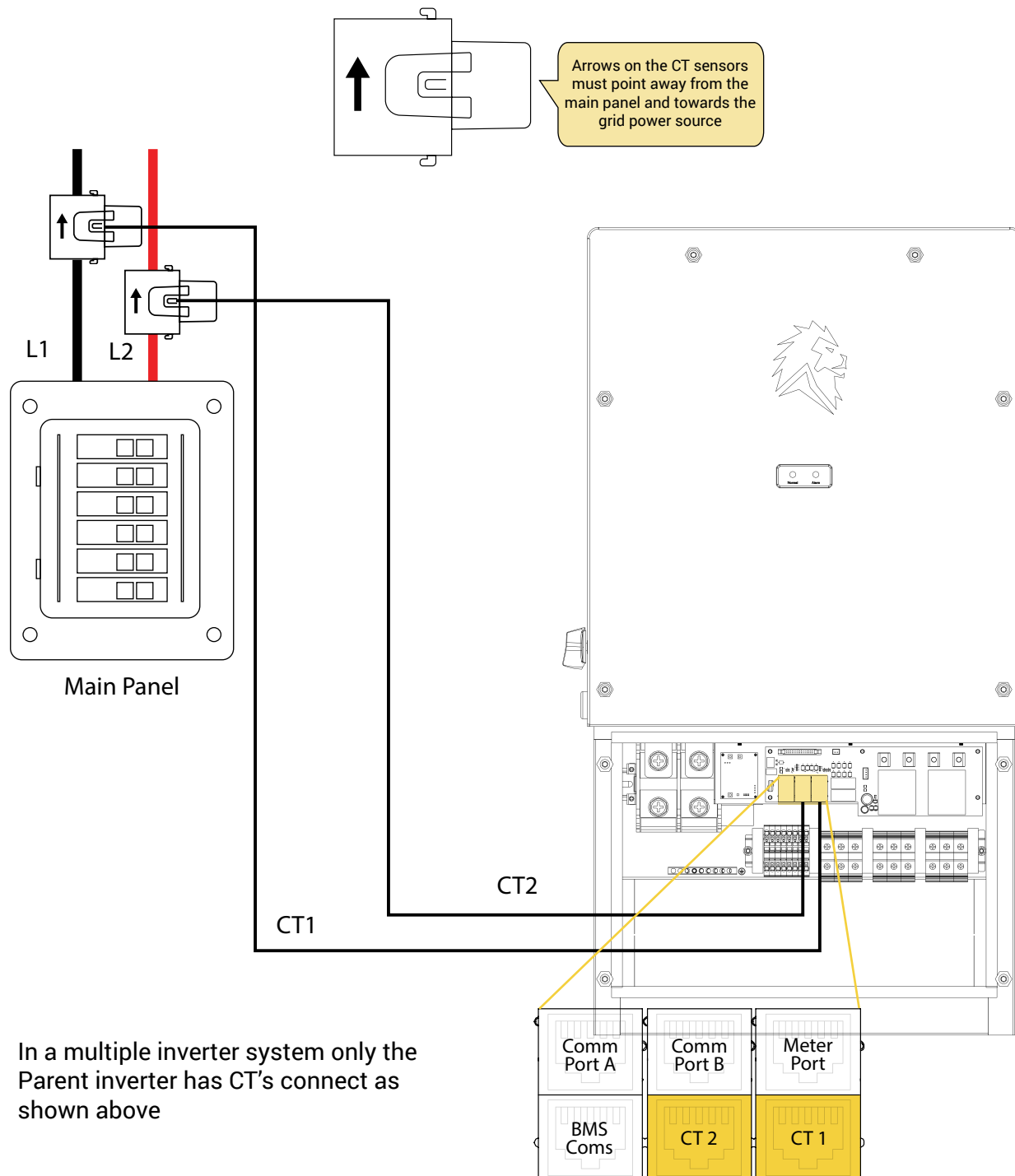
With power still off, measure continuity between load L1 and load L2 of each inverter. It should be open circuit on each inverter.

If any of these tests don't pass, the wiring needs to be corrected before proceeding.

Turn off the AC output breakers. During commissioning, you will be prompted to reconnect and check.

CT Installation

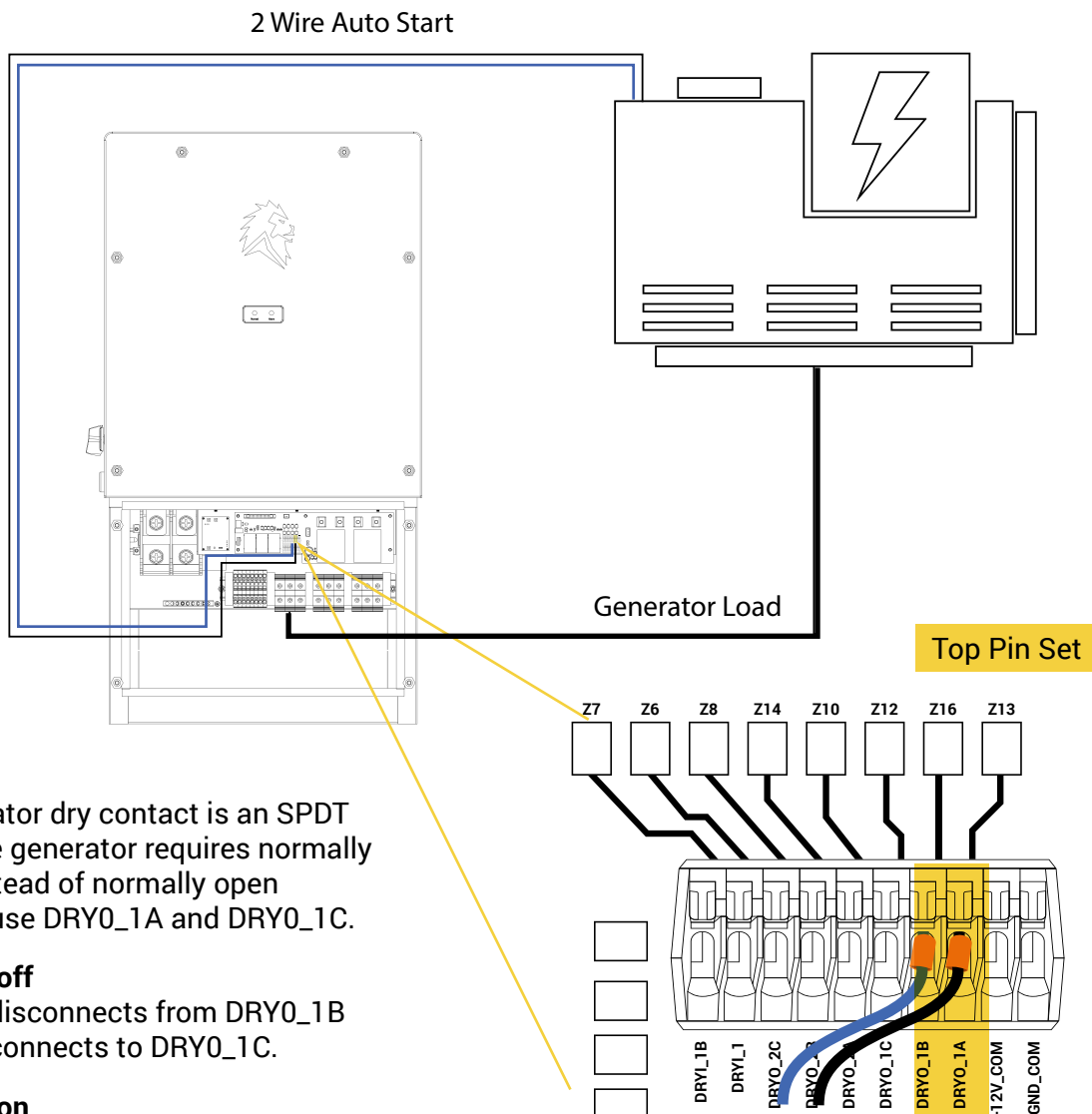
Using the provided CT's, clamp the CT loops in front of the main service panel with the arrow on the CT pointed away from the Sanctuary ESS towards the grid. Ideally, the CT's should be placed on the lines from the utility meter so that the entire home consumption is measured.



Generator Wiring : 1 Inverter

The Sanctuary ESS is compatible with 240V split phase home backup generators. The Sanctuary can function with both a manual or an auto start feature. It can accept up to 8kW of input power through the Gen input to cover loads and to charge batteries. The generator functions are only available when the grid is not present.

To wire a generator to the Sanctuary ESS, the generator AC output will go directly into the Gen input of the Sanctuary inverter. If the generator has a remote two wire auto start feature, these lines must be wired into the DRYO_1B and DRYO_1A ports on the inverter circuit board.



The generator dry contact is an SPDT relay. If the generator requires normally closed instead of normally open contacts, use DRYO_1A and DRYO_1C.

Generator off

DRYO_1A disconnects from DRYO_1B
DRYO_1A connects to DRYO_1C.

Generator on

DRYO_1A connects to DRYO_1B
DRYO_1A disconnects from DRYO_1C.

Generator Wiring: Multiple Inverter



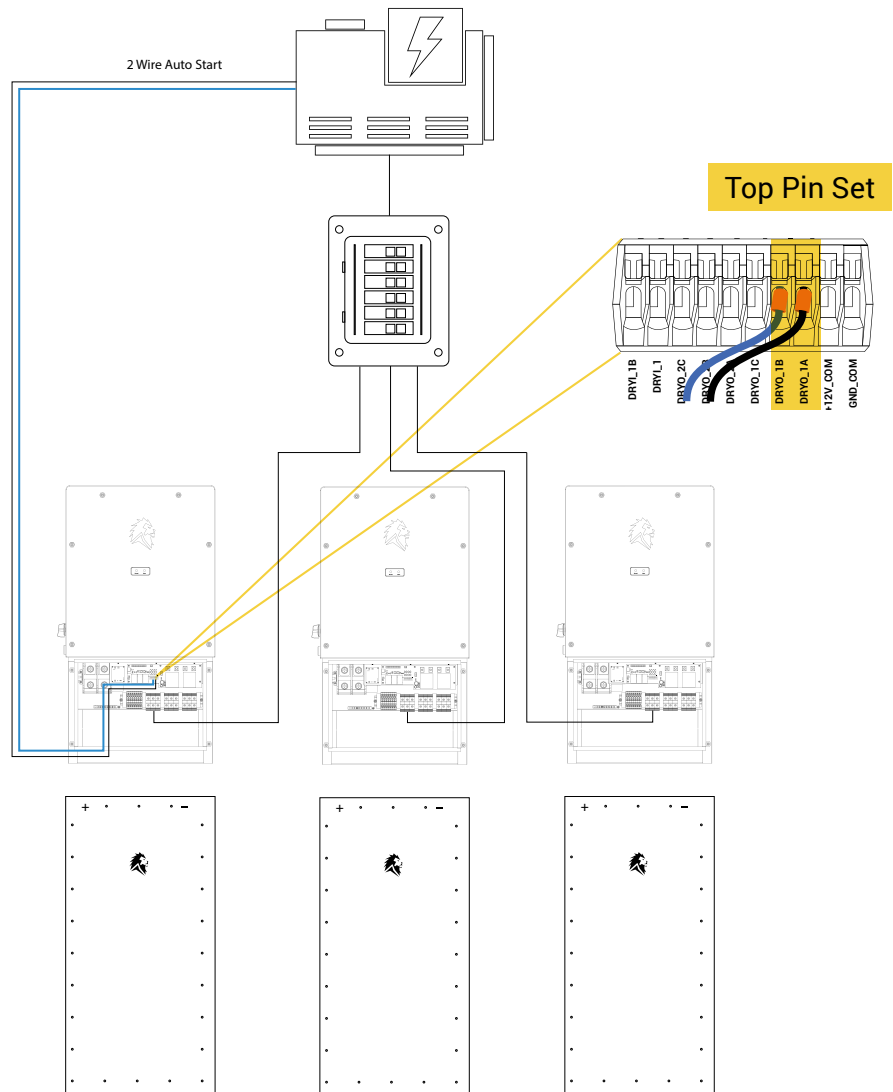
L1 L2 PHASING

It is of vital importance that the grid input and load output phases on multiple breakers match. Crossing phases WILL cause a catastrophic failure on the equipment. See generator continuity testing on page 33.

The Sanctuary ESS is compatible with 240V Split Phase home backup Generators. The Sanctuary can function with both a manual or an auto start feature. It can accept up to 8kW of input power through the Gen Input to cover loads and to charge batteries. The Generator function is only available when the grid is not present.

When the install consists of multiple inverters, the generator AC output needs to be wired to a combiner panel. From the combiner panel, lines will go to each Sanctuary Inverter's Gen Inputs. If the generator has a remote two wire auto start feature, these lines must be wired into the Parent inverter's DRYO_1B and DRYO_1A inputs.

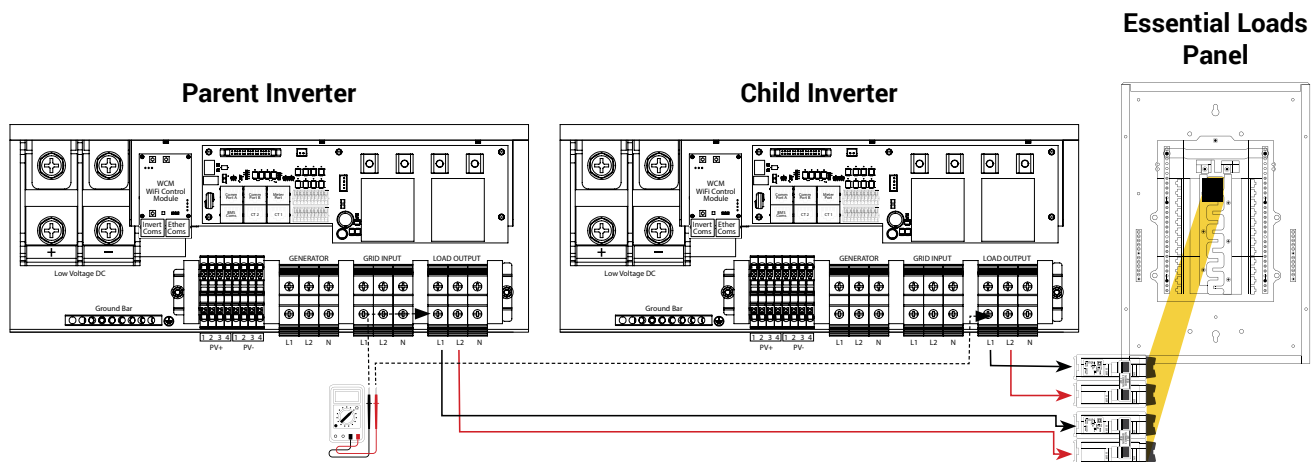
***** Any time multiple inverters are installed, a continuity test must be performed to ensure all phases match before powering up.**



Breaker Sizing

The three breakers feeding the sanctuary system from the generator combiner panel need to be the same amp rating with a maximum of 40A each.

Generator Input Continuity Testing

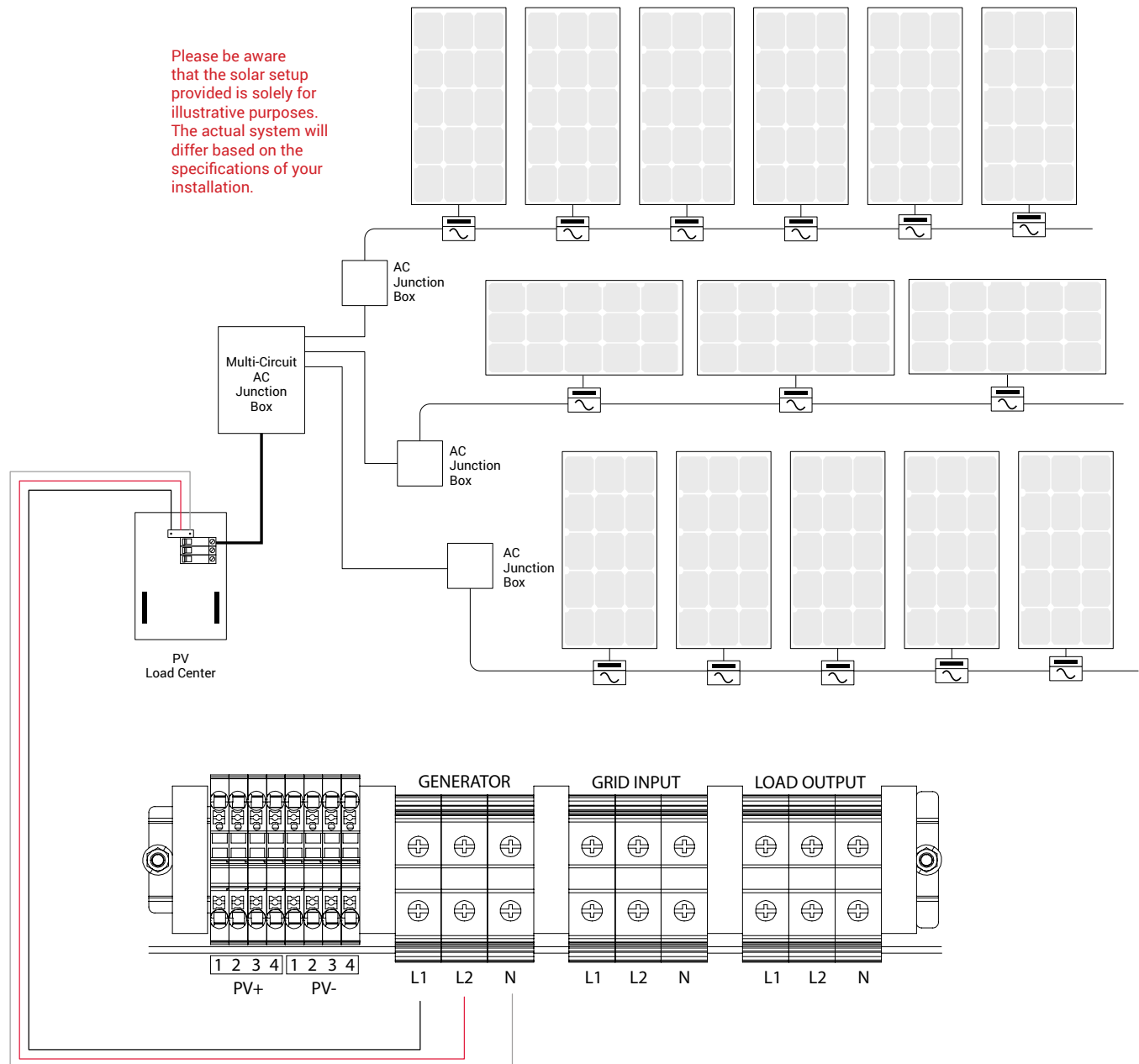


- Step 1.** Make sure the generator is off. If using AC solar, Make sure the AC power button on each inverter is turned off. If the inverter is turned on with AC solar enabled, the generator port will be connected to grid power through internal relays.
- Step 2.** Leave the generator (or AC solar) combiner breakers in the on position.
- Step 3.** Make sure the power is off by measuring voltage between L1 and L2 on each inverter's generator port.
- Step 4.** Check continuity between generator L1 on the first inverter and generator L1 on the second inverter. These should be close to zero ohms. Generator L1 on the second inverter to generator L1 on the third inverter should also be close to zero ohms.
- Step 5.** Check continuity between generator L2 on the first inverter and generator L2 on the second inverter. These should be close to zero ohms. Generator L2 on the second inverter to generator L2 on the third inverter should also be close to zero ohms.
- Step 6.** Turn off the breaker on the generator. Leave the breakers on that feed the generator port of each inverter. Measure continuity between the generator L1 and generator L2 ports on the inverter. It should be open circuit

AC Solar

This is not recommended for new installations since it's not as efficient, but it is a convenient option for existing grid-tied systems.

- Connect AC solar to generator port

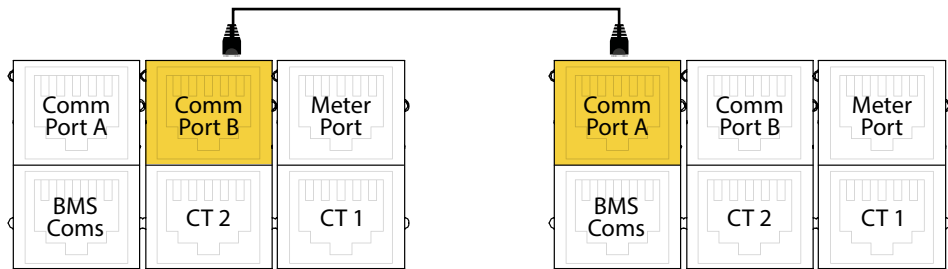
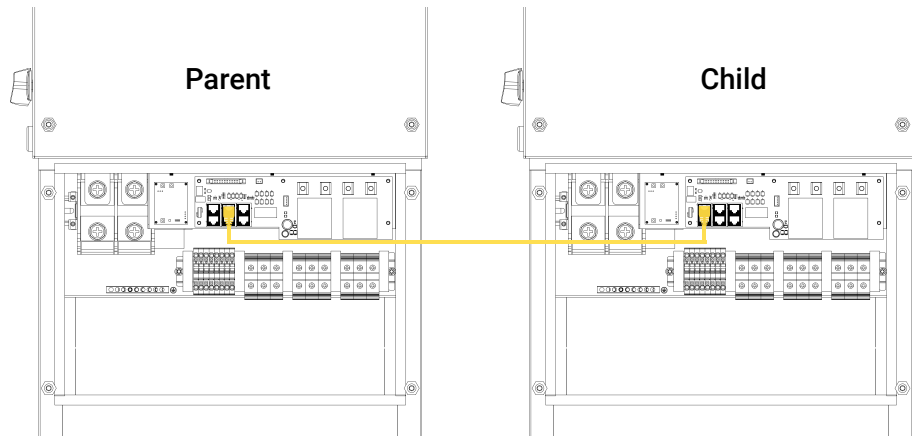


Parallel Inverter Connection

When wiring 2 inverters in parallel, using the provided cat5 cable:

Connect cat5 cable on Parallel B located on the Parent Inverter to Parallel A on the child Inverter as shown.

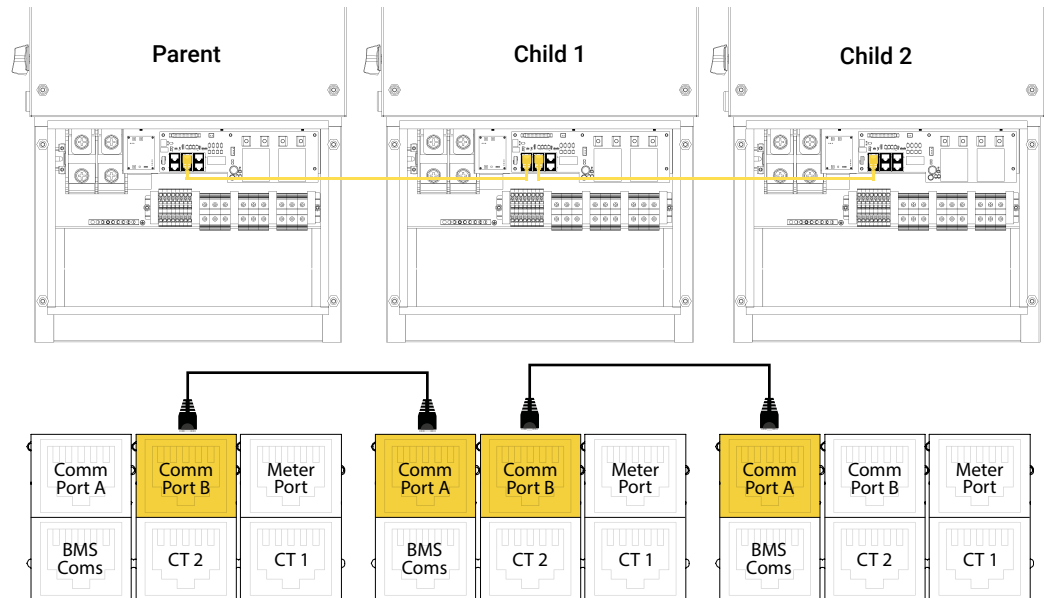
We recommend using shielded Ethernet cables.



When wiring 3 inverters in parallel, using the provided cat5 cables:

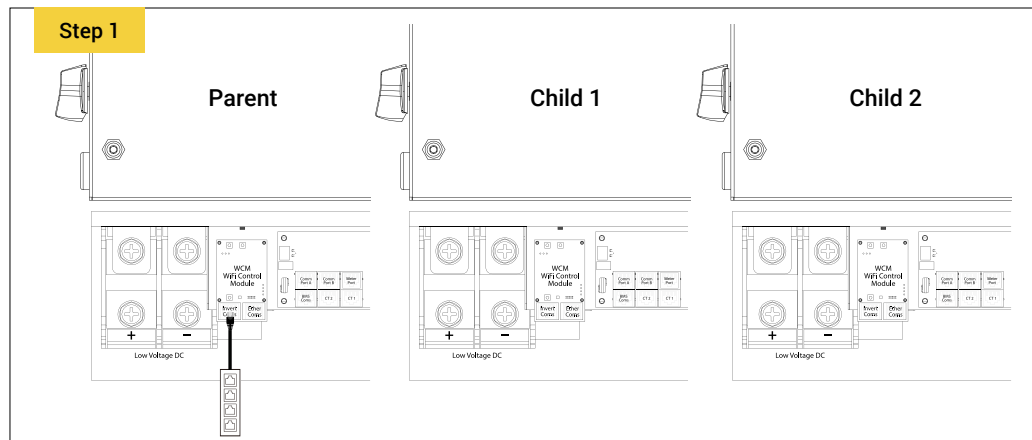
Connect the first cat5 cable on Parallel B located on the Parent Inverter to Parallel A on the Child 1 Inverter as shown.

Next, connect a second Cat5 cable from Parallel B on Child 1 to Parallel A on Child 2.

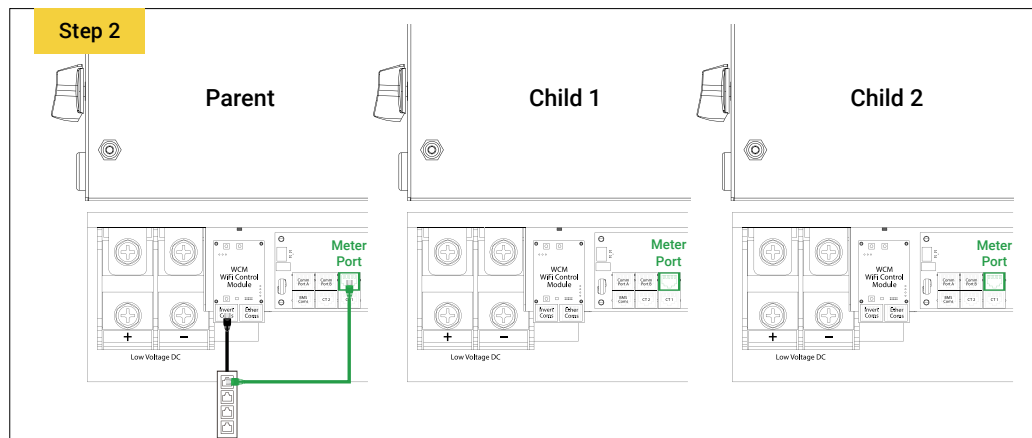


Inverter Communication (Comm) Wiring

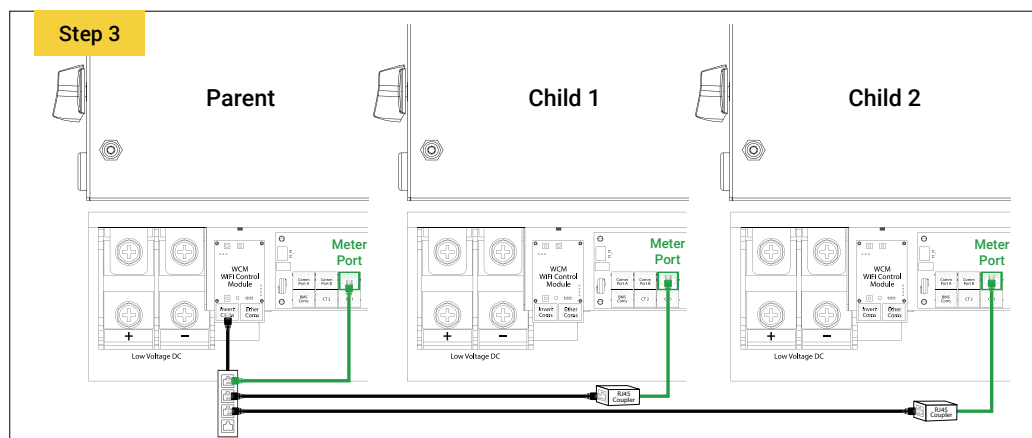
Step 1: On the Parent Inverter, connect the provided 4-port RJ45 splitter to the WCM comm port.



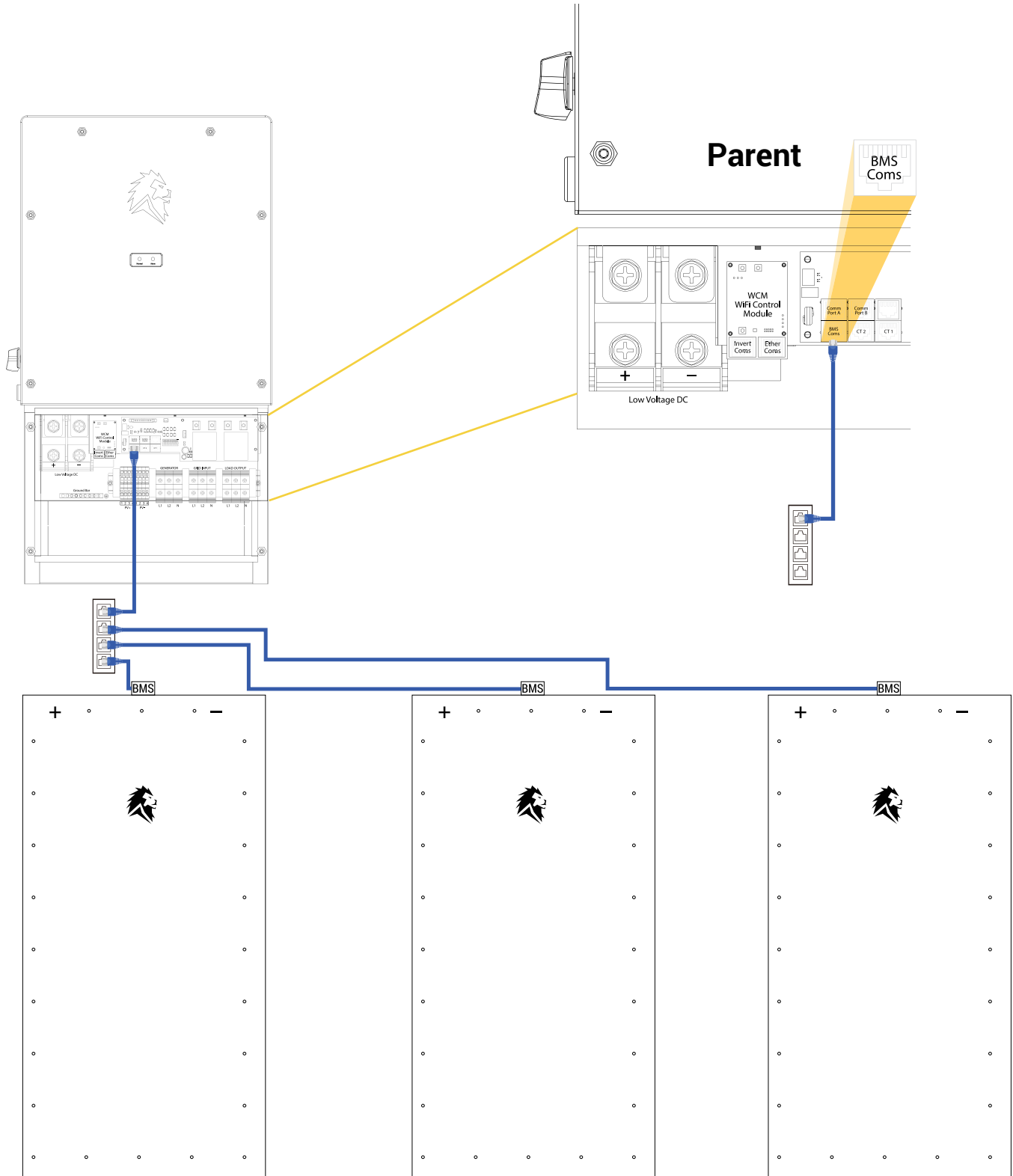
Step 2: Connect the provided 10" flat cable from the RJ45 splitter to the meter port on the inverter.



Step 3: On the Child inverters, connect one end of the provided 10' cat5 patch cables to the coupler connecting to the 10" flat cable connected to the meter port. Connect the other end of the 10' cat5 patch cable to the 4-port RJ45 splitter

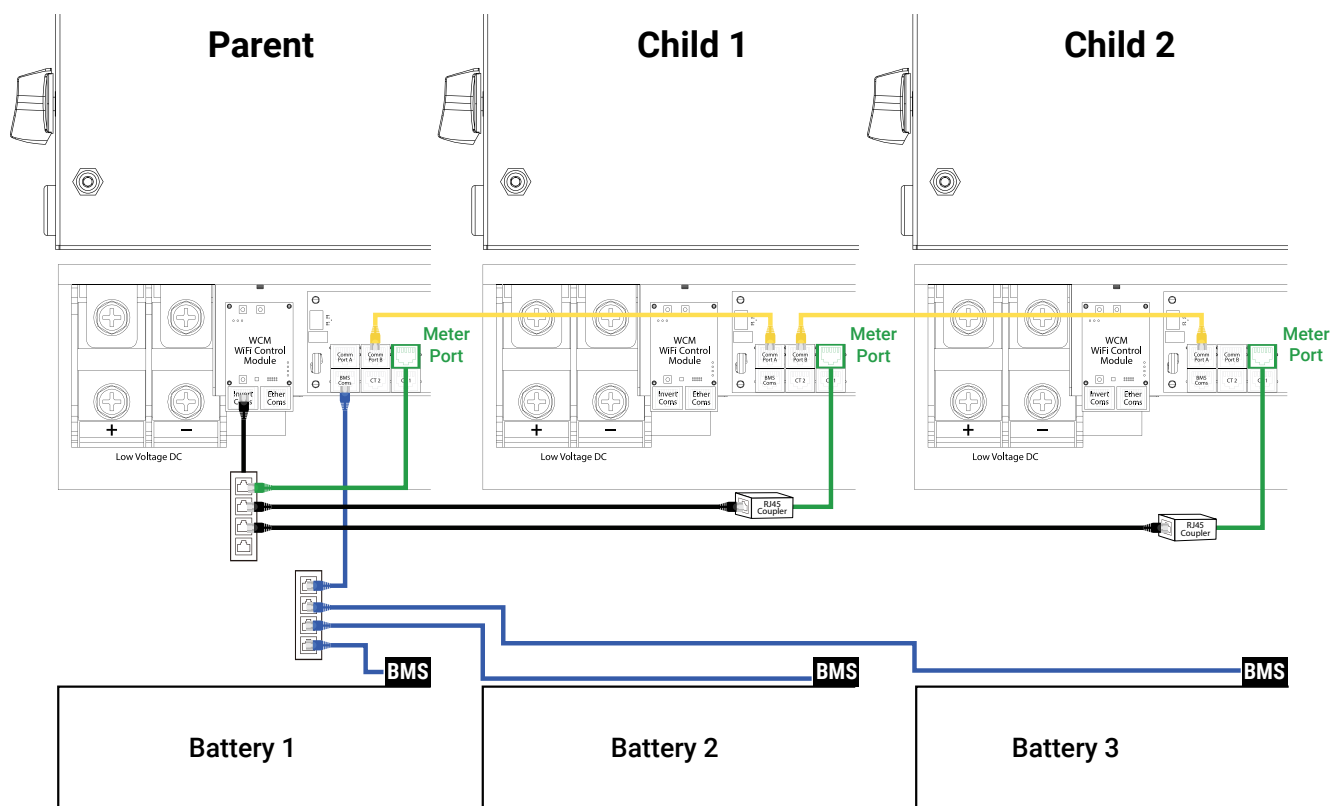


BMS Communication Cable Wiring



Inverter Wiring Diagram

- Parallel Communication Wiring
- Communication Wiring
- BMS Communication Cable Wiring



You're now ready to begin the commissioning process. Please make sure you have completed the training and have received a login to the Lion Energy App. Please visit smart.lionenergy.com

Sanctuary 12kW Energy Storage Inverter



Shown is the 99990814 13.5kWh Version

CAUTION:

Read all instructions before installation, operation and maintenance of the system

Install in a climate controlled environment to maintain a temperature between 32° to 86° F

This Grid Support Interactive Inverter complies with ULSTD. 1741, UL1741 SB, IEEE Std 1547:2018, IEEE Std 1547.1:2020, FCC 15 class-B, UL1699B Arc-Fault Circuit



SANCTUARY™ ENERGY STORAGE SYSTEM

Model #99990814 – 12K Hybrid Solar Inverter with 13.5kWh Energy Storage

Model #99990815 – 12K Hybrid Solar Inverter with 27.0kWh Energy Storage

Model #99990816 – 12K Hybrid Solar Inverter with 40.5kWh Energy Storage

Model #50170168 Hybrid Solar Grid Support Interactive Inverter

PV Input

Max PV Input	12kW
MPPT Input Voltage	120-500V
Max Input Current	12A*4
Max Short Circuit Current	15A

AC Output (On Grid)

Rated Output Power	8kVA
Rated Output Current	33.3A
Grid Voltage	120V/240V Split phase
Grid Frequency (Optional)	60Hz
Power Factor Range	-0.8~+0.8

AC Output (EPS)

Rated Output Power	8kVA
Rated Output Current	33.3A
EPS Voltage	120V/240V Split phase
EPS Frequency (Optional)	60Hz

Battery

Battery Voltage Range	40V~60V
Max Charging Current	190A
Max Discharging Current	190A

System

Ingress Protection	IP65
Dimensions	430*710*220mm
Weight	41Kg
Max Efficiency	97.8%
AC Solar Max	8kW
Grid Passthrough	90A
Voltage Range	40 - 55.6VDC
Capacity	13,875.2Wh
Parallelable Capacity	41,625.6Wh Max
Quantity	3 Max
Charging Temperature Range / Current	32° to 86° F / 150A
Discharging Temperature Range / Current	-4° to 86° F / 160A
Weight	277lbs.
Dimensions	39.75" H x 18.5" W x 10"D
Compliance	UL1973, UL9540, UL9540A



Lion Energy Sanctuary 10-Year Warranty

The Lion Sanctuary 10-year limited warranty ("Warranty") applies to manufacturing defects and workmanship. The warranty begins from the date the Sanctuary System is installed. System must be connected to the internet and Lion Energy must be granted monitoring access in order to qualify for warranty.

If the unit needs to be repaired or replaced, follow these steps:

1. Customer contacts the Installer to verify there is an issue with the unit.
2. Installer contacts Lion Energy to discuss the issues and Lion Energy will determine if the unit will be repaired or replaced.
3. Installer repairs the unit or if the unit is to be replaced, Lion Energy will ship the unit to the Installer.
4. Installer replaces the unit and ships the replaced unit to Lion Energy for proper recycling.

The warranty does not include:

- Damage during transportation of equipment.
- Damage caused by incorrect installation or commissioning.
- Damage caused by failure to comply with operation instructions, installation instructions, or maintenance instructions, including but not limited to Lion Energy prescribed firmware updates.
- Damage caused by attempts to modify, alter, or repair products.
- Damage caused by incorrect use or operation.
- Damage caused by operating the battery outside its allowable voltage range.
- Damage caused by draining battery below the maximum depth of discharge allowed by Lion Energy, which is limited to 90% of total battery capacity.
- Damage caused by operating the battery above the maximum charge and discharge power limits set within the Sanctuary Energy Storage System Manual.
- Damage caused by insufficient ventilation of equipment.
- Damage caused by failure to comply with applicable safety standards or regulations.
- Damage caused by natural disasters or force majeure (such as floods, lightning, over voltage, storms, and fires).
- Any external scratches will not affect the basic operation of the product. Any external scratches, stains or natural mechanical wear does not represent a defect in the product.

Battery Configuration	Energy Retention ¹	Operating Limitation
1 Battery	70% at 10 years following initial installation date	41.1 MWh of aggregate throughput ²
2 Battery	70% at 10 years following initial installation date	82.2 MWh of aggregate throughput ²
3 Battery	70% at 10 years following initial installation date	123.3 MWh of aggregate throughput ²

¹ Expressed as a % of the useable battery rated capacity

² Measured at the battery DC output

Limitation of Liability

THIS LIMITED WARRANTY IS THE ENTIRE WARRANTY AND ANY OTHER EXPRESSED OR IMPLIED WARRANTIES ARE NOT APPLICABLE. THIS WARRANTY EXCLUDES ANY LIABILITY FOR PRODUCT NOT BEING AVAILABLE FOR USE OR LOST REVENUES OR PROFITS. Lion Energy only warrants the product, not the power service or external electrical equipment or any services provided by another party. This manual provides information, specification, and usage instructions. All statements, information, and suggestions within in this manual do not constitute any express or implied warranty.

Recycling

Return the Sanctuary System to Lion Energy for proper disposal.



Lion Energy Sanctuary 25-Year Warranty

The Lion Sanctuary 25-year limited warranty ("Warranty") applies to manufacturing defects and workmanship. The warranty begins from the date the Sanctuary System is installed. System must be connected to the internet and Lion Energy must be granted monitoring access in order to qualify for warranty.

If the unit needs to be repaired or replaced, follow these steps:

1. Customer contacts the Installer to verify there is an issue with the unit.
2. Installer contacts Lion Energy to discuss the issues and Lion Energy will determine if the unit will be repaired or replaced.
3. Installer repairs the unit or if the unit is to be replaced, Lion Energy will ship the unit to the Installer.
4. Installer replaces the unit and ships the replaced unit to Lion Energy for proper recycling.

The warranty does not include:

- Damage during transportation of equipment.
- Damage caused by incorrect installation or commissioning.
- Damage caused by failure to comply with operation instructions, installation instructions, or maintenance instructions, including but not limited to Lion Energy prescribed firmware updates.
- Damage caused by attempts to modify, alter, or repair products.
- Damage caused by incorrect use or operation.
- Damage caused by operating the battery outside its allowable voltage range.
- Damage caused by draining battery below the maximum depth of discharge allowed by Lion Energy, which is limited to 90% of total battery capacity.
- Damage caused by operating the battery above the maximum charge and discharge power limits set within the Sanctuary Energy Storage System Manual.
- Damage caused by insufficient ventilation of equipment.
- Damage caused by failure to comply with applicable safety standards or regulations.
- Damage caused by natural disasters or force majeure (such as floods, lightning, over voltage, storms, and fires).
- Any external scratches will not affect the basic operation of the product. Any external scratches, stains or natural mechanical wear does not represent a defect in the product.

Battery Configuration	Energy Retention ¹	Operating Limitation
1 Battery	50% at 25 years following initial installation date	63.5 MWh of aggregate throughput ²
2 Battery	50% at 25 years following initial installation date	127 MWh of aggregate throughput ²
3 Battery	50% at 25 years following initial installation date	190.5 MWh of aggregate throughput ²

¹ Expressed as a % of the useable battery rated capacity

² Measured at the battery DC output

Limitation of Liability

THIS LIMITED WARRANTY IS THE ENTIRE WARRANTY AND ANY OTHER EXPRESSED OR IMPLIED WARRANTIES ARE NOT APPLICABLE. THIS WARRANTY EXCLUDES ANY LIABILITY FOR PRODUCT NOT BEING AVAILABLE FOR USE OR LOST REVENUES OR PROFITS. Lion Energy only warrants the product, not the power service or external electrical equipment or any services provided by another party. This manual provides information, specification, and usage instructions. All statements, information, and suggestions within in this manual do not constitute any express or implied warranty.

Recycling

Return the Sanctuary System to Lion Energy for proper disposal.

Warranty Program



Thanks for choosing the Lion Sanctuary Lithium Energy Storage System. It is a high-quality, hybrid inverter and energy storage unit that can be expandable. It comes with a standard 10 year warranty and can be upgraded to a 25 year warranty.

To activate the warranty, fill out all the information below and return it to Lion Energy. All the information is required. You can scan it or send a photo of this page to Warranty@LionEnergy.com

Warranty Information

Inverter Serial Number (on Side Panel): _____

Battery #1 Serial Number (on Side Panel): _____

Battery #2 Serial Number (on Side Panel): _____

Battery #3 Serial Number (on Side Panel): _____

Installation Date: _____

Physical Address of Installation: _____

Location in the home of Installation: _____

Name of Installation Company and Installer: _____

Phone Number of Installation Company: _____

Number of Solar Panels at Location: _____

Brand of Solar Panels: _____

Solar Panel Model #: _____

VOC # (On Panels): _____ PVM # (On Panels): _____ Total PV on Site: _____

Check Applicable Warranty: ☐ Standard 10 Year ☐ Upgraded 25 Year



385.375.8191
Monday - Friday
9:00 AM - 5:00 PM MST



735 S. Auto Mall Drive Suite 200
American Fork, Utah
84003



warranty@lionenergy.com
www.lionenergy.com
[@lionenergy](https://www.lionenergy.com)

Notes:

[illegible]



LIONENERGY™



385.375.8191
Monday - Friday
9:00 AM - 5:00 PM MST



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